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Modelling resources to supply Ethiopia with renewable electricity by 2030

FRIDA JALKENÄS

MONIKA MIZGALEWICZ



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Frida Jalkenäs, Monika Mizgalewicz

Department of Sustainable Energy Technology, Royal Institute of Technology, Stockholm, Sweden

Bachelor of Science Thesis

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A Minor Field Study (MFS) conducted in Addis Ababa, Ethiopia, spring 2017

Supervisor: Mark Howells, mark.howells@energy.kth.se

Energy Technology Department, Division of Energy System Analysis, KTH, ITM

Examinator: Cecilia Sundberg, cecilia.sundberg@abe.kth.se

Industrial Ecology and Environmental Management Department, KTH, ABE



This study has been carried out within the framework of the Minor Field Studies Scholarship Programme, MFS, which is funded by the Swedish International Development Cooperation Agency, Sida.

The MFS Scholarship Programme offers Swedish university students an opportunity to carry out two months' field work, usually the student's final degree project, in a country in Africa, Asia or Latin America. The results of the work are presented in an MFS report which is also the student's Bachelor or Master of Science Thesis. Minor Field Studies are primarily conducted within subject areas of importance from a development perspective and in a country where Swedish international cooperation is ongoing.

The main purpose of the MFS Programme is to enhance Swedish university students' knowledge and understanding of these countries and their problems and opportunities. MFS should provide the student with initial experience of conditions in such a country. The overall goals are to widen the Swedish human resources cadre for engagement in international development cooperation as well as to promote scientific exchange between universities, research institutes and similar authorities as well as NGOs in developing countries and in Sweden.

The International Relations Office at KTH the Royal Institute of Technology, Stockholm, Sweden, administers the MFS Programme within engineering and applied natural sciences.

Erika Svensson
Programme Officer
MFS Programme, KTH International Relations Office

Abstract

Energy is a crucial factor when it comes to development. Among other electricity is important when advancing the living standards of a society as it facilitates various actions and mechanisms. Through the past years social trends such as increased population have put energy and electricity systems under stress as they have often been based on limited and unsustainable fossil fuels. A need for a shift from the conventional fuels to renewable sources becomes more prominent and development needs to be performed in a sustainable way.

Ethiopia is one of the countries who have expressed a desire to reach sustainable development by adapting United Nations' Sustainable Development Goals. This project focuses and analyzes more specifically Ethiopia's relation to, and possibility of reaching Goal 7 - sustainable energy for all. It attempts to find a future configuration of energy sources which will create an electricity system which will benefit the economy, the social aspect and the environment. The aim is to have a cost-efficient energy mix which supplies all of Ethiopia's inhabitants with electricity without having to contribute with any carbon dioxide emissions.

A literature review is performed to obtain country specific information such as geographical predispositions, and a field trip to Addis Ababa is conducted where data regarding the electricity system is collected. Modelling is then carried through by usage of tools MoManI and OnSSET and the obtained results show a continuous trend in all scenarios where solar and wind compromise the biggest part of electricity production in 2030 and after. Every scenario also allows all inhabitants access to electricity by year 2030. Further, four out of five scenarios ensures elimination of carbon dioxide emissions by 2022, and all five by 2030.

Keywords: CLEWs, Electricity, Energy, Ethiopia, OnSSET, OSeMOSYS, Sustainable development

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Supervisor's testimonial

Working with Frida and Monika has been a great experience. They both showed an interest in the topic, high motivation, critical thinking and persistence despite the hurdles encountered. Consequently, their output (both models and report) is of high quality and very well structured. If the grading scale used was A-E, this work would receive a well deserving A.

Georgios Avgerinopoulos

PhD Candidate, KTH

georgios.avgerinopoulos@desa.kth.se

0046 - 73 567 20 37

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List of Abbreviations

Abbreviation	Stands for
BAU	Business as usual
CLEWs	Climate, Land, Energy and Water systems
CSP	Concentrated Solar Power
EEP	Ethiopian Electric Power
EEU	Ethiopian Electric Utility
GHG	Greenhouse gas
GLPK	GNU linear programming kit
I	Refers to entities that, in the modelling base year, 2010, already have access to electricity. This includes industry, transport, agriculture, residential, commercial and public services
KTH-dESA	The division for Energy Systems Analysis at the Royal Institute of Technology
MoManI	Model Management Infrastructure for OSeMOSYS
OnSSET	Open Source Spatial Electrification Toolkit

OSeMOSYS	Open Source Energy Modeling System
PP	Power plant
R	Refers to the population, in rural areas, who gains electricity access gradually after 2017
RES	Reference Energy System
SDG	Sustainable Development Goal
SHS	Solar Home System
TEMBA	The Electricity Model Base for Africa
U	Refers to the population, in urban areas, who gains electricity access gradually after 2017
UN DESA	United Nations Department of Economics and Social Affairs

1. Introduction

Energy is a service that has become a prerequisite for the society to be able to function and develop. However, due to trends such as growing population, improved living standards and urbanization the demand increases significantly, and conventional energy sources, such as fossil fuels, become queried due to their depletable nature, limited reserves as well as their effects on the environment.

Various actions are taken in order to tackle the energy dilemma described above, and instead of just development, focus is put onto *sustainable* development, which was defined by the World Commission on Environment and Development as “[...] development that meets the needs of the present without compromising the ability of future generations to meet their own needs.” (World Commission on Environment and Development, 1987). The United Nations have been working with sustainable development for many years. In 2000 they presented the Millennium Development Goals and following their deadline in 2015, Sustainable Development Goals were created (UNDPa, n.d.). The Sustainable Development Goals (SDGs) address amongst other poverty termination, protection of the environment and establishment of peace and prosperity (UNDPa, n.d.).

This project is based on an analysis of the energy system of Ethiopia and thus, particular attention is brought to SDG 7 - Affordable and Clean Energy. The goal emphasizes the importance of access to electricity, concurrently highlighting the dangers associated with an economy reliant on fossil fuels. Its main objective is to ensure access to affordable, reliable, sustainable and modern clean energy for all (UNDPb, n.d.). Although focus is put on SDG 7, all the SDGs are interconnected and improvement in one area may bring progress in other areas as well. For example, electricity is a requirement for many manufacturing processes, especially those occurring on larger scales. Power outages can hinder production, delaying the process or even stopping it completely, which causes economic losses. Stable and reliable access to electricity is thus a key aspect of economic growth (SDG 8 – Decent work and economic growth). Furthermore, it reassures entrepreneurs and encourages innovation that boosts the industry (SDG 9 - Industry, innovation and infrastructure).

Additionally, having electricity in a household indicates access to light, which facilitates various activities, for example studying (SDG 4 - Quality Education). It may also simplify tasks such as cleaning and cooking. As these activities are often performed by women, easing them will give women more free time that they can use to educate themselves or go to work which will reduce gender inequalities (SDG 5 – Gender equality). Further, using clean energy reduces various toxic emissions which implements improved life in water and on land (SDG 14 - Life below water, SDG 15 – Life on land). It also contributes to SDG 13 – Climate Action.

An analytical framework that deals with the integration of and synergies between various resource systems is CLEWs - Climate, Land-Use, Energy and Water strategies. This project will target climate and energy, analysing the electricity sector in Ethiopia. Data collected in Ethiopia by interviews and studies of national policies, will be used to investigate the country’s potential to become fully electrified and carbon free by 2030. This will be done by the use of the modelling tools OSeMOSYS and OnSSET.

1.2 CLEWs

CLEWs (Climate, Land-Use, Energy and Water strategies) is an integrated approach used in studies related to - among other - sustainable development. It focuses on resources within the sectors of climate, land, energy and water and the objective is to identify interlinkages and trade-offs between the different sectors. This is done in order to detect areas where synergies occur and where utilization of resources can be made more effective. An extensive connection exists for example between land, energy and water - resources strongly linked to people’s wellbeing. Water plays an essential role in land-usage and energy systems, energy is needed for water supply and land-usage, and land is required to create energy and water infrastructure (Department of Energy Technology, 2015).

Furthermore, the use of the land-, energy- and water resources can affect climate change and microclimate, with the extent depending on how and to what degree they are exploited. Climate change, on the other hand, affects the systems that provide these resources thus creating a linkage of mutual influence (Howells et al., 2013).

An integrated system approach is used in the CLEWs nexus, it investigates interconnections between the resource sectors in order to determine counterintuitive feedbacks and the effect a change in one sector can have on another (Department of Energy Technology, 2015).

1.3 Overview of Ethiopia and its electricity sector

Ethiopia is located in Eastern Africa, has an area of 1 104 300 km² and borders 6 countries namely: Sudan, South Sudan, Somalia, Kenya, Djibouti and Eritrea. With a population of approximately 99,4 million people, it is the most populous landlocked country in the world (CIA, 2017).

In 2014 no less than 73 million of the population lived without access to electricity. In the rural areas only 10% had access to electricity and the national electrification rate was 25% which, compared to the global electrification rate of 84%, could be considered low (IEA, 2014). In areas where electricity is already available it is characterized by instability, resulting in frequent power outages. This occurs even in the capital city, Addis Ababa (Müggenburg et al., 2011), resulting in a reliance on private, off-grid and fossil fuel driven generators to produce electricity (UNDP, 2011). The reason behind the power outages may vary but are partially due to the fact that the government sometimes prioritizes to export energy to neighbouring countries in order to get foreign currency, rather than supplying its own population¹. Another common cause is due to a distribution system of poor quality.

On top of this, the currently installed generation capacity is 2,261 MW (USAID, 2017), which entails that the average amount of electricity available to each person is considerably below the global standards. The current electricity generation mix consists of 86% hydroelectric, 8% out of other renewables and 6% out of thermal energy (USAID, 2017). The high share of renewables results in the fact that the power sector only accounts for 3% of the total GHG emissions (UNDP, 2011).

Ethiopia is one of the fastest growing non-oil producing economies in Africa and depends highly on its agricultural sector, which accounts for 80% of the country's total employment (UNDPc, n.d.). Up to this day, the economic growth has also been driven mostly by progress in the agricultural and service sector and has been achieved while maintaining the lowest level of income inequality in Africa (UNDPc, n.d.). However, despite the fast-growing economy Ethiopia remains one of the poorest countries in the world (CIA, 2017).

In 2015 19,5% of the population lived in urban areas and the average annual urban population growth rate between 2010 and 2015 was 4,9%. During the same period, the average population growth for the whole country was 2,5% (UNdata, 2017) which indicates a rapid growth, that puts a lot of pressure on land and water resources, and results in an increased demand for electricity (CIA, 2017).

Ethiopia's climate is described by the Central Intelligence Agency (CIA) as tropical monsoon. Differences in topography contribute to varying climate across the country (CIA, 2017). The average rainfall per month varies greatly throughout the year; in 2012 the least being 9.54 mm in January and the greatest being 124.43 mm in July (The World Bank, 2012). Fluctuations in rainfall affect greatly the mainly rain-fed agriculture and thus also the economy. The anthropogenic climate change and additional climate extremes can thus be devastating for the country (Conway & Schipper, 2011). Ethiopia can already see the effects of climate change with an increased average temperature and changes in rainfall patterns (UNDP, 2011). One clear example is that the country, in 2016, suffered from their worst drought

¹ Azagew, Sileshi; Head of the Infrastructure Department, National Planning Commission; interview on the 5th of April 2017, Addis Ababa

in 50 years, resulting in more than 18 million people in the need of food aid (Laing, 2016).

Additionally, unstable precipitation also affects the electricity production that is majorly based on hydropower. Today Ethiopia uses about 5% of its total, 45 000 MW hydro potential (Asnake & EEP, 2015). In order to minimize the effects of droughts on the energy sector Ethiopia stated in their Intended Nationally Determined Contribution (INDC) that efforts will be put into expanding electric power generation from geothermal, wind and solar sources (Federal Democratic Republic of Ethiopia, 2016). These actions will also decrease the competitiveness in the water, energy and food nexus.

2. Research question and objectives

The purpose of this project is to provide insights into the expansion of the electricity sector of Ethiopia. Furthermore, the aim is to perform a first assessment of Ethiopia's climate, land, energy and water nexus, in a way that could help the country achieve the Sustainable Development Goals.

The objective is to obtain relevant information about Ethiopia's electricity system today as well as future outlooks, and then create relevant scenarios to use as a basis for energy modelling. The goal is to find a configuration of the energy system which will benefit the economy, the society and the environment.

The research question of this project is: *Which energy mix, in Ethiopia, ensures electricity for all inhabitants and no carbon emissions by 2030, at the lowest cost? What is the amount of investments needed to reach SDG 7?*

3. Methodology

This study is divided into two main parts - literature review and data collection, as well as energy modelling. The first phase of the project consists of getting an overview of Ethiopia and understanding social trends taking place in the country, such as population and economic growth. More detailed knowledge is obtained regarding the core of this project, that is the energy system in Ethiopia. Information concerning the current energy mix and climate is collected from sources such as CIA's The World Factbook, UNdata and the World Bank.

Data from TEMBA (see section 3.1.5) is given by KTH-dESA and put into the models. A business as usual scenario is created for Ethiopia in MoManI (interface for running the OSeMOSYS code). A reference energy system (RES) is also drawn based on the TEMBA-data (see figure 1). This gives an overview of the system that is going to be modelled and allows for a better understanding of the connections between the different parameters. The RES serves as a schematic representation of the real energy system in Ethiopia and shows which fuels and technologies are available and have potential in the given area. Technologies represent units that transform something from one form to another and fuels are a direct form of energy. In the RES, technologies are presented as boxes and fuels as lines. The RES also represents the boundaries of the study.

Thereafter a search for data which could be the foundation for determining scenarios follows. This takes place partially in Sweden but mainly in Ethiopia. Through interviews with contacts at the National Planning Commission of Ethiopia and other companies, and internet- and archive based searches information is gained regarding national policies - current and future ones. Useful documents about goals and strategies with planned investments, energy demand projections as well as forecasts about growth rate regarding the population and economy are obtained.

Based on the retrieved information, different scenarios for the modelling are constructed. The guidelines when choosing and designing the scenarios are primarily Ethiopia's national goals and UN's Sustainable Development Goal 7. For more detailed information about how the scenarios were chosen

and developed, see appendix II and III.

Results are then compiled and presented in diagrams and tables. This is followed by an analysis and discussion. Finally, the main points are highlighted in the conclusion.

3.1 Modelling tools

This study focuses on system cost optimization. The modelling is conducted using the following tools.

3.1.1 Open Source Energy Modelling System tool, OSeMOSYS

OSeMOSYS uses a linear programming method to optimize the system. Given the chosen constraints (natural, technical and political) it finds the most cost-efficient energy mix to meet the demand. It is written in GNUMathprog, which is a free and open source software, and uses the GNU linear programming kit (GLPK) solver (Howells et al., 2011). To run the OSeMOSYS code, different interfaces can be used, in the case of this project, a software called MoManI is used.

3.1.2 Model Management Infrastructure interface, MoManI

MoManI is an open source interface for energy modelling. It was developed by UNDESA and KTH-dESA in order to facilitate the use of OSeMOSYS (Department of Energy Technology, 2016).

The initial modelling set-up used for this project is the following: the total energy demand is divided into three different sectors; I, U and R. I represents the demand coming from entities that, in the modelling base year, already have an established access to electricity. U and R represent those who gain electricity access as the years in the modelled lifespan progress. U covers the urban areas while R considers the rural areas (KTH-dESA, 2017). The modelled period in MoManI stretches from 2010 to 2040.

3.1.3 Open Source Spatial Electrification Tool, OnSSET

OnSSET is an optimization energy modelling tool that estimates the most cost-efficient electrification option. Unlike other energy modelling tools, OnSSET considers geographical data related to energy, such as population density, the consumer's distance to the grid and road networks, and night-time light. The results can thereby show where the grid should be extended and where it's better to have mini-grid (i.e. a local grid detached from the national one) or stand-alone systems. With input data regarding costs for grid extensions and different technologies it quantifies the needed investments for increasing electrification in the given area. The off-grid renewable solutions include solar photovoltaics, wind turbines as well as mini and small hydropower plants. The electrification rate in the base year assumes no off-grid solutions (Mentis et al., 2016). The modelled period in OnSSET stretches from 2015-2030, but the results are only shown for the end year, 2030.

3.1.4 OSeMOSYS-OnSSET integration

Firstly, a harmonization of data is performed. This includes for an example an addition of mini grid and stand-alone technologies in OSeMOSYS so that both model consider the same technologies. The demand is also adjusted accordingly. Subsequently, an estimation of the Grid Cost with primary results from OSeMOSYS is performed. The Grid Cost is then used as an input in OnSSET. The output from OnSSET is, on the other hand, information regarding the off-grid technology split and is used as an input in OSeMOSYS. Iterations are performed until no significant changes are observed.

3.1.5 The Electricity Model Base for Africa, TEMBA

With data from the World Bank the electricity supply in each African country is modelled and connected to each other by trade links in a model called TEMBA. It is built in OSeMOSYS to facilitate national energy planning and is the world's first model of African electricity supply (Taliotis et al., 2016). The

development was sponsored by the World Bank and SIDA, and is constantly under modification (KTH-dESA, n.d.). The business as usual (BAU) scenario created in this project is based on data from TEMBA (see appendix IV).

The TEMBA data assumes that the electrification rate will be 100 % by 2030 and that each household corresponds to 5 people. The energy intensity for both rural and urban population in Ethiopia is set to be 696 kWh/household in 2030 (KTH-dESA, n.d.).

3.2 Study boundaries

A reference energy system (RES) has been created in order to obtain a better overview of the energy system, see figure 1. The RES is a representation of how the model sees Ethiopia. It gives an overview of the parameters and their connections to each other.

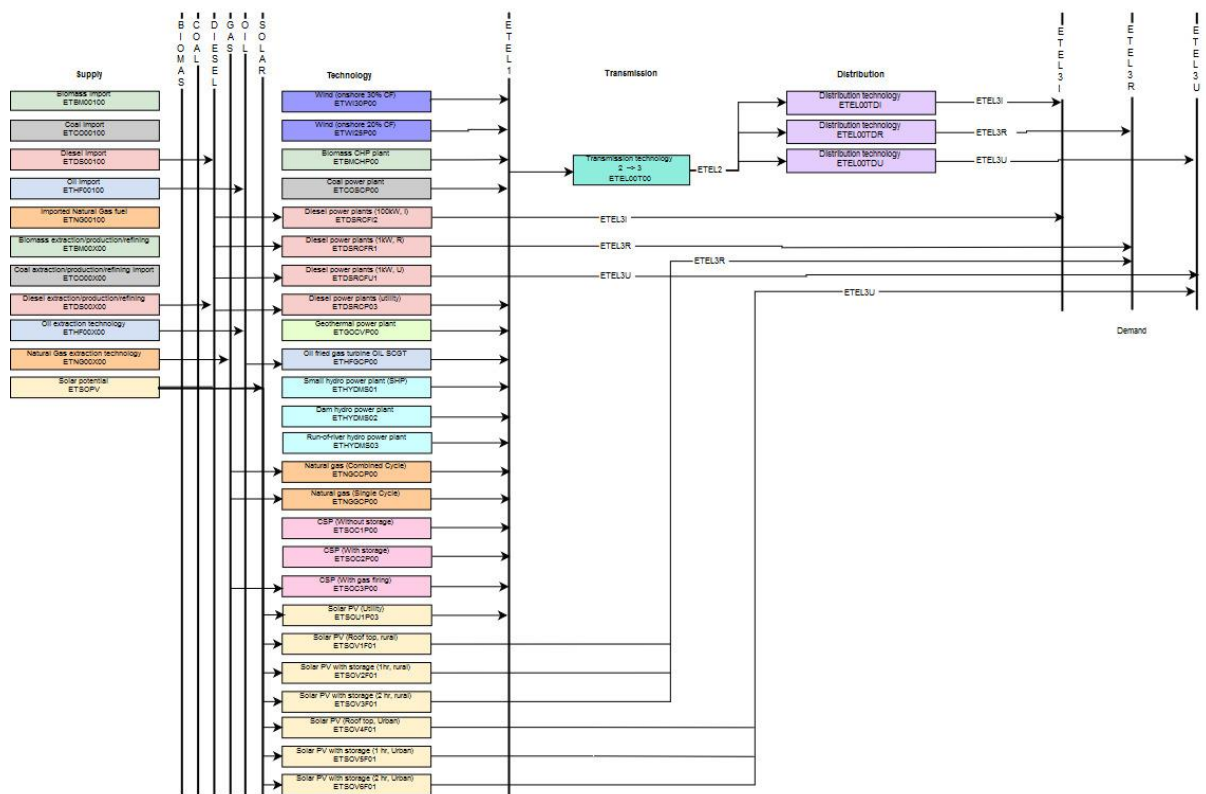


Figure 1: Reference Energy System, RES

Furthermore, this project is constrained to only consider electricity. Thus, throughout this report, when energy is mentioned no other forms of energy than electricity are taken into account. This means that for example heat, transportation and non-electrified cooking are excluded, which leads to a simplified demand profile that does not correspond to the reality. The reason for this constraint is limited time and lack of data.

The models created in MoManI do not take into account the energy losses that occurs in the transmission and distribution system. When it comes to GHG emissions, other emissions than CO₂ are excluded. No trade with external parts is considered either.

Additionally, OnSSET is limited to consider only activities which cannot increase an area's productivity, i.e residential demand, so for example it includes a private household's electricity requirements but not a hospital's, a school's or an industry's.

3.3 Chosen scenarios

Below is a brief description of the modelled scenarios and a short reasoning as to why they were chosen. For more detailed information on how they were constructed, see appendix II and III.

3.3.1 Scenario A

Scenario A is the business as usual, BAU-scenario and was created based on data from TEMBA. Alterations were made to the distribution and transmission parameters. This is due to the fact that MoManI assumes there is no distribution or transmission system prior to the base year of the modelling and creates those parameters by itself as the production of the energy starts. This is not the actual case as there is already a quite extensive distribution and transmission established in the country. The alterations to the standard data were conducted by taking the output value for AccumulatedNewCapacity in 2010 (after running the model once) for both the distribution- and the transmission technology, entering it as an input for the ResidualCapacity for each technology and running again.

3.3.2 Scenario B

The main objective in scenario B is to improve the distribution and transmission system, which according to Sileshi Azagew from the National Planning Commission² is lacking and a big contributor to the many power outages in Ethiopia. The ones who are connected to the grid should be able to rely on the national supply and get access to electricity throughout the day without the need of using of a generator. A reliable connection is also crucial to factories and businesses, and attracts investors from the outside too. One way to implement the scenario where improvement is achieved is by making the model invest more in distribution and transmission. This is done in scenario B by entering a decreasing residual capacity for the distribution and transmission systems. A decreasing value will force the model to do investments in the current systems, which, by assumption, lead to improvements.

3.3.3 Scenario C

The Ethiopian government has many ambitious plans for the future development of the energy system. Scenario C was therefore created in order to see the result of the current plans and strategies (see 1.3.1 Relevant national policies). A residual capacity was entered for the already existing technologies based on their capacity and the model was forced to invest in different technologies according to the plans in GTP II and the Master Plan (see appendix I).

3.3.4 Scenario D

This scenario was created in the same way as scenario C, but with a higher demand. The country is in an economical situation where they depend strongly on export to other countries, in order to get foreign currency into the country. With its high potential for renewable energy Ethiopia has a strong possibility to export low carbon energy to the neighbouring countries. A higher export could lead to an increased economic growth which in turn could lead to improved living standards followed by an increased energy demand. The demand in this scenario is therefore based on a forecast to 2030 that was presented in the CRGE (see appendix I).

3.3.5 Scenario E

If Ethiopia shall reach the SDG 7 by 2030 it is interesting to know where the grid should be extended and where it is more cost efficient to supply people with electricity by mini grid or a stand-alone system.

² Azagew, Sileshi; Head of the Infrastructure Department, National Planning Commission; interview on the 5th of April 2017, Addis Ababa

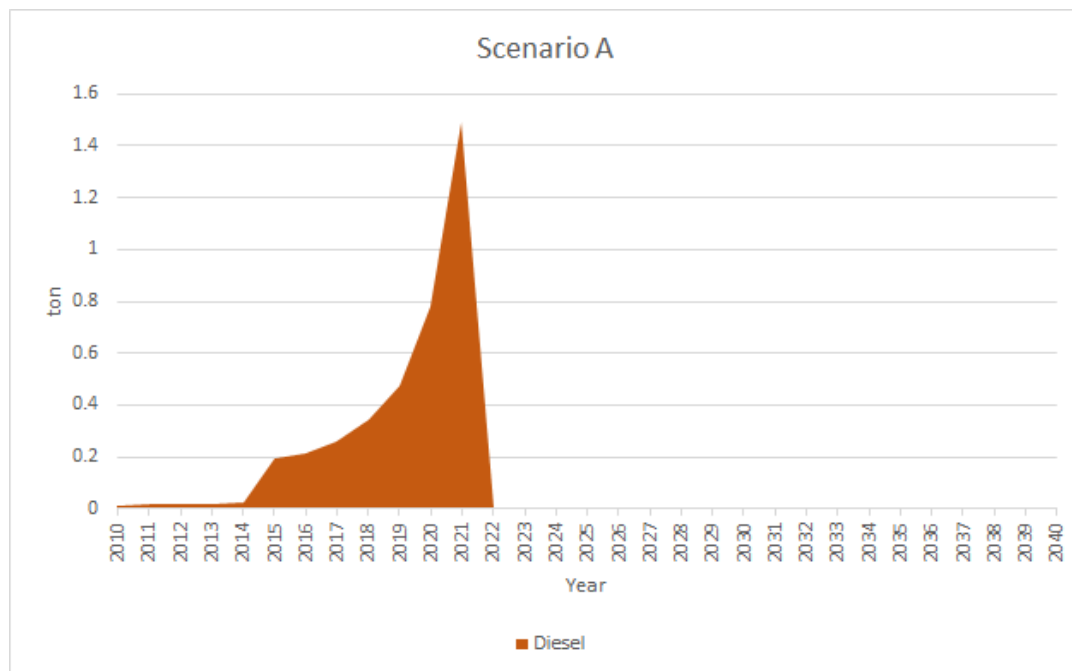
To get these results an integration between OSeMOSYS and OnSSET was conducted. The integration was chosen to be done with a clone of scenario C since this scenario represents the national goals and a demand without big exports.

4. Results

The most significant changes are presented in figures below and a discussion is followed in the next section. In the figures technologies connected directly to the grid are named only by the source, ex. *wind*. If the technology is stand alone or mini grid, it is indicated by the legend.

Note that for the following results a bug was detected in the code after the report was submitted. The model itself is correct but the coding, which is not in the scope of this project, is flawed. This causes all hydro to disappear after a period, around year 2031. This might create some confusion as it is natural to think that with growing demand it would be more cost-efficient to run in already existent hydro power plants than to build new facilities. As the results are consequences of a defective code a further evaluation of the results is required. The model and subsequently the report are planned to be revised.

4.1 Annual Emissions



The annual emissions from 2010 to 2040 are presented in the figures below.

Figure 2: Annual emissions, scenario A

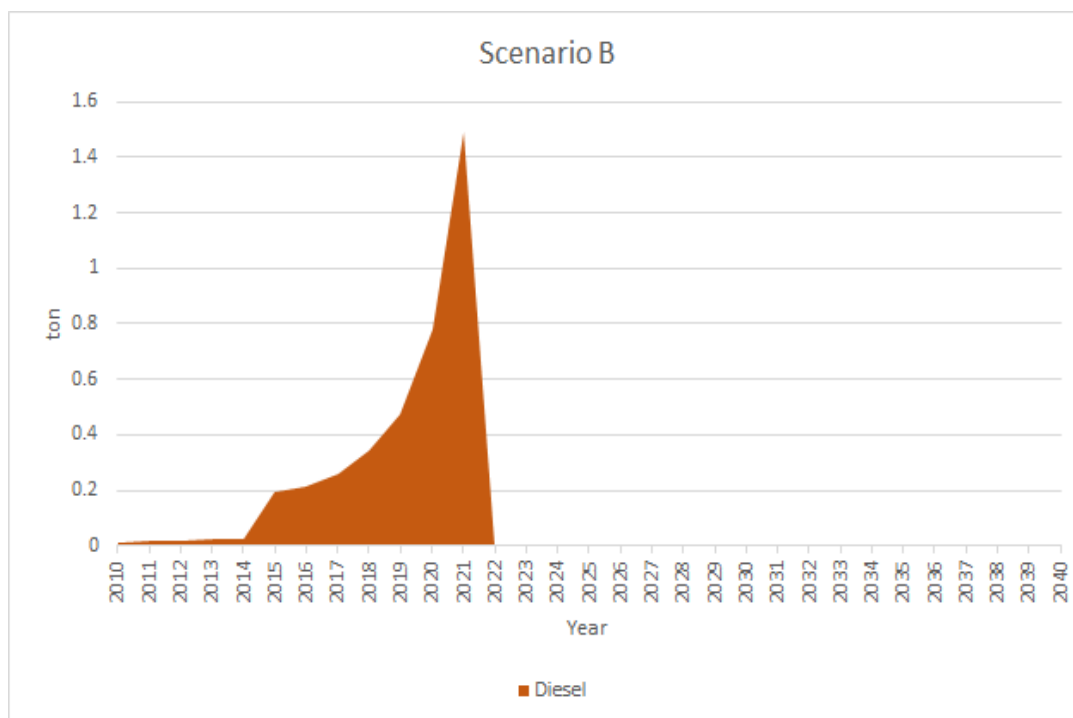


Figure 3: Annual emissions, scenario B

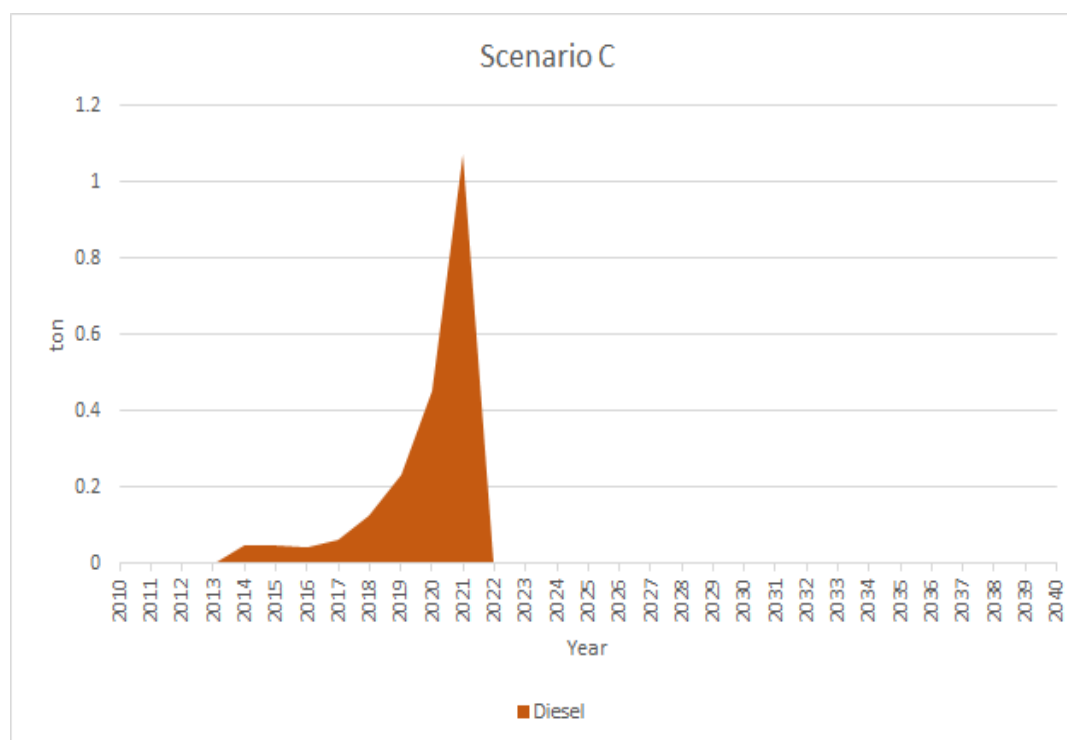


Figure 4: Annual emissions, scenario C

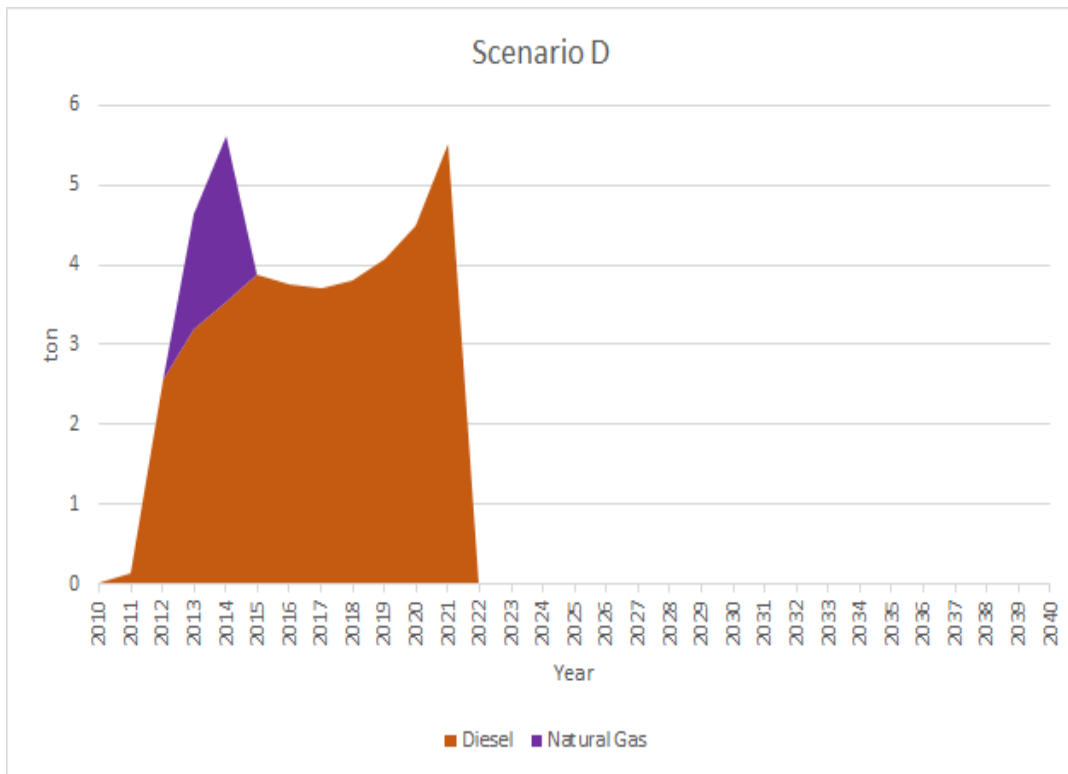


Figure 5: Annual emissions, scenario D

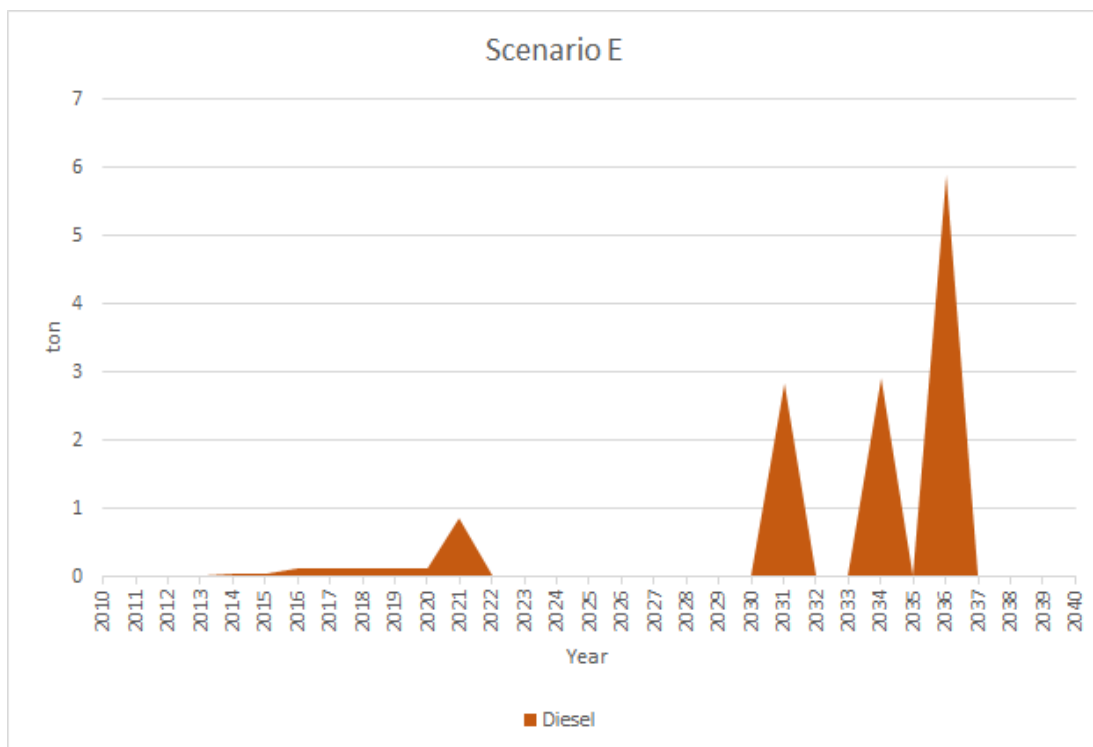


Figure 6: Annual emissions, scenario E

4.2 Capital Investments

Below the total capital investments are presented for the modelled scenarios. Figures are presented with scenario specific data as well as one where all scenarios are compared to each other.

Table 1: Capital Investment, 2010-2040

Scenario	A	B	C	D	E
Capital investment in distribution and transmission (M\$)	27 141,41	27 712,03	27 141,41	34 113,63	18 181,26
Total capital investment (M\$)	147 438,1	141 534	173 839,52	230 423,10	243 825,78

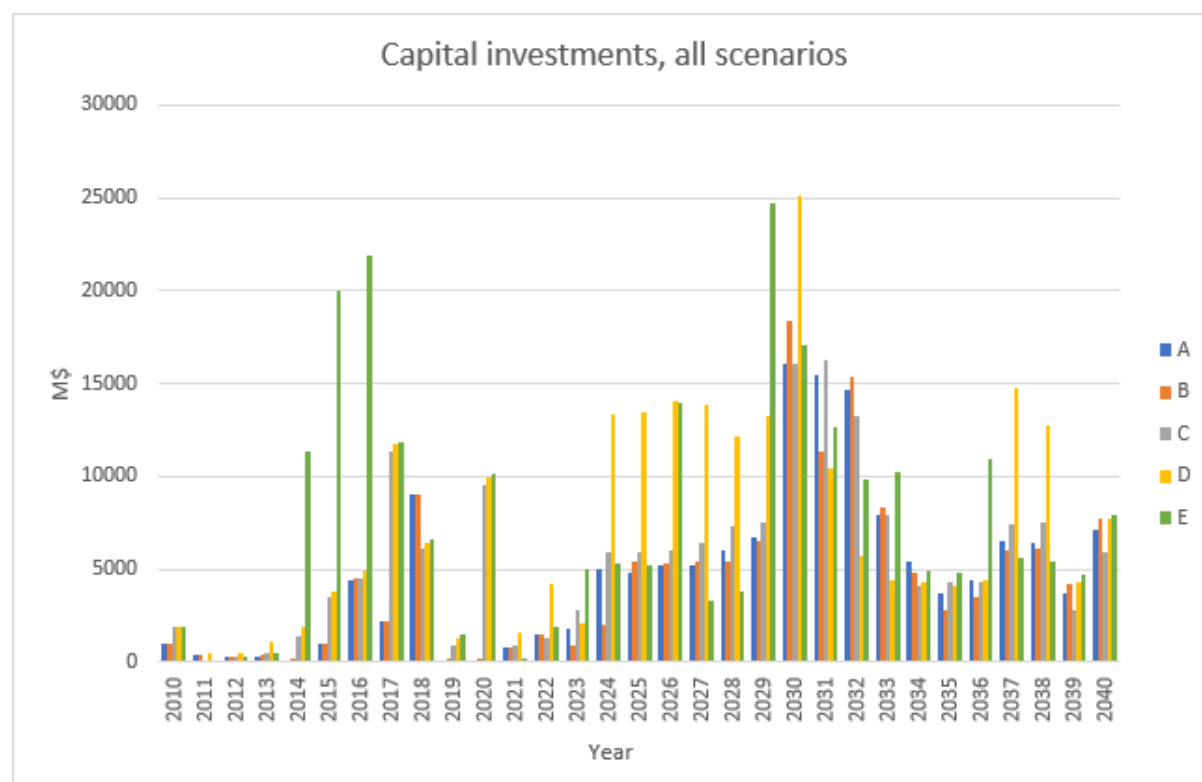


Figure 7: Capital Investment by year for each scenario

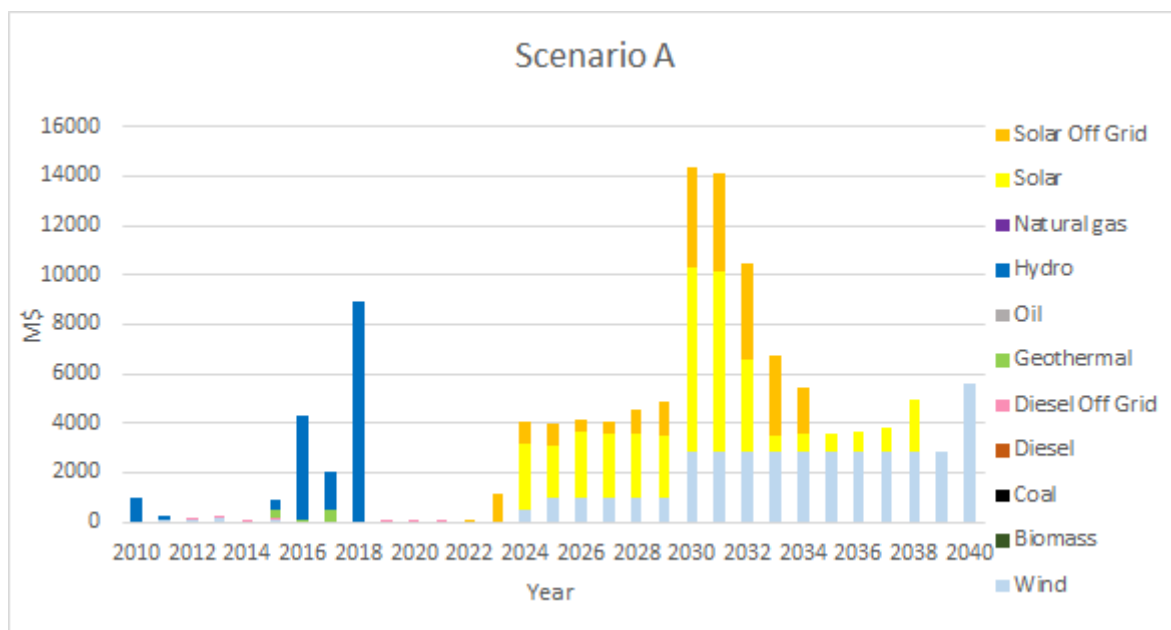


Figure 8: Capital Investment by year for scenario A

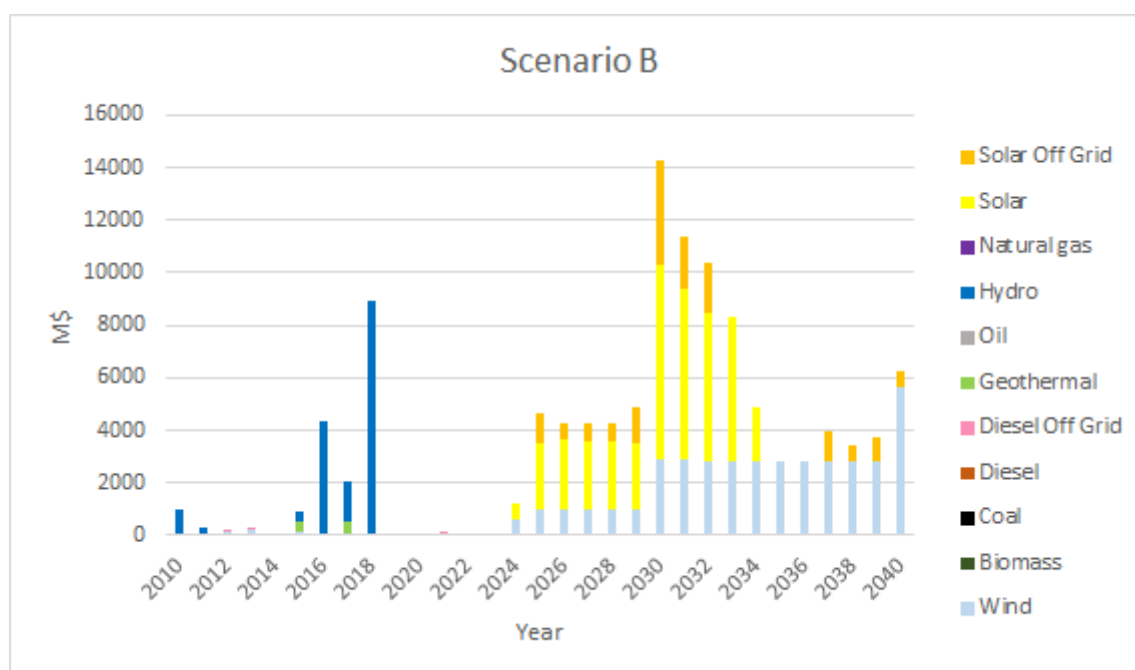


Figure 9: Capital Investment by year for scenario B

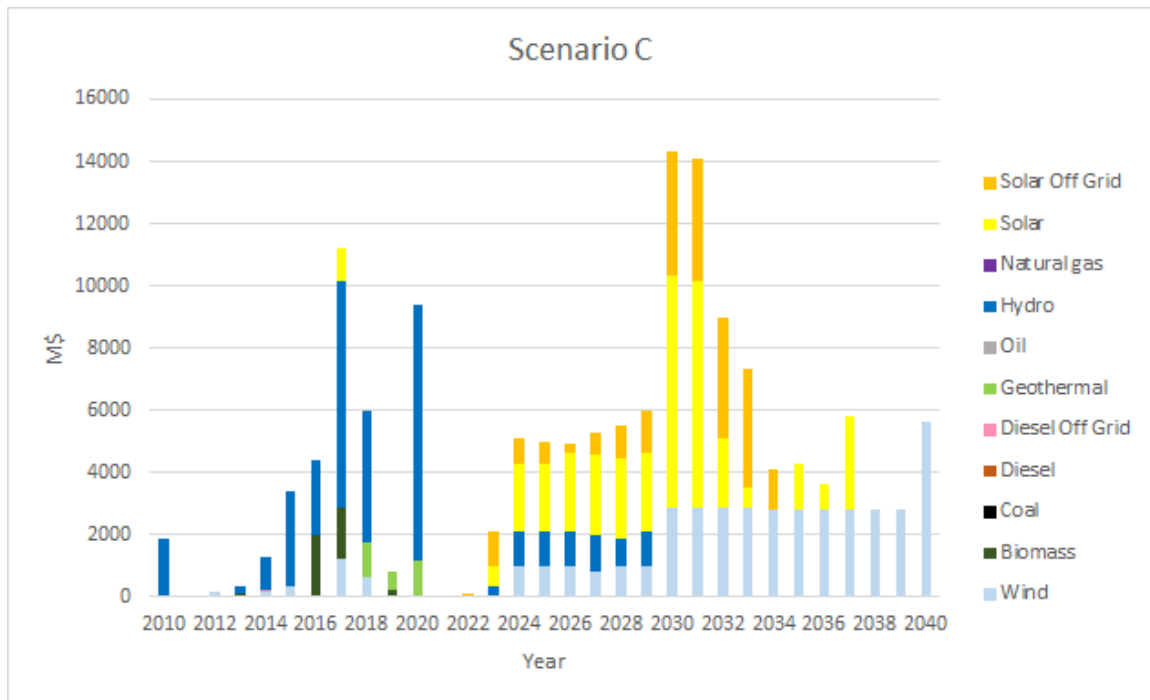


Figure 10: Capital Investment by year for scenario C

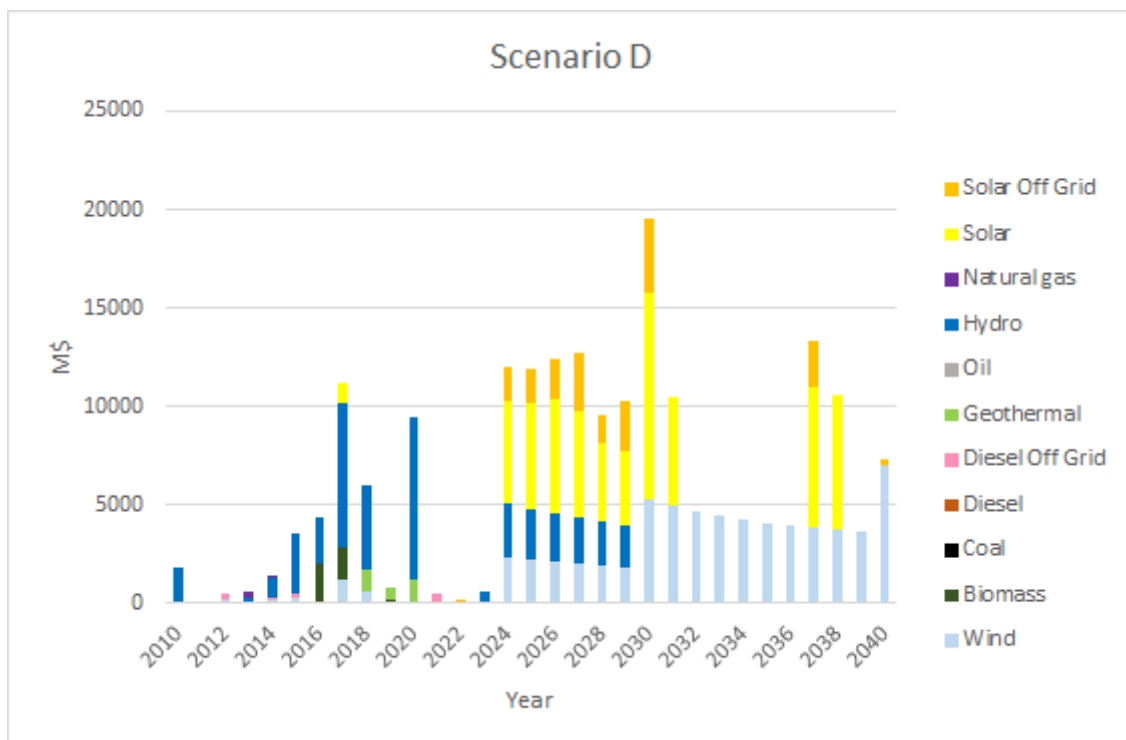


Figure 11: Capital Investment by year for scenario D

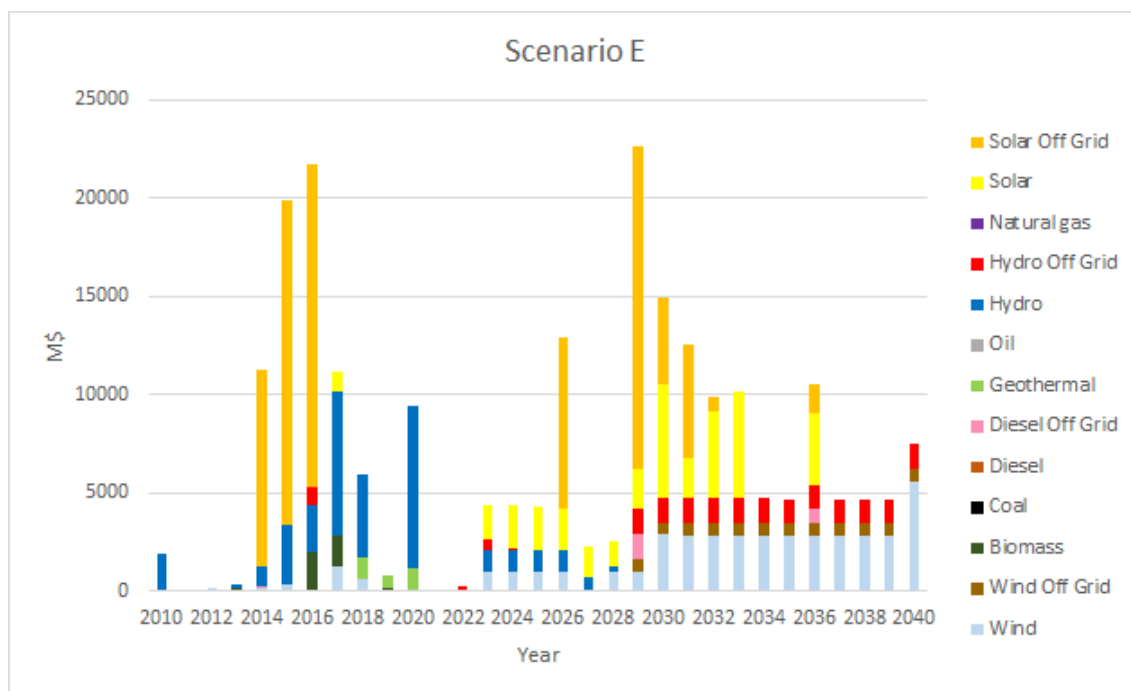


Figure 12: Capital Investment by year for scenario E

4.3 Production by Technology Annual

Figures with the annual production by technology for each scenario are presented below. A comparison between the different scenarios and their production by 2030 is presented in table 2.

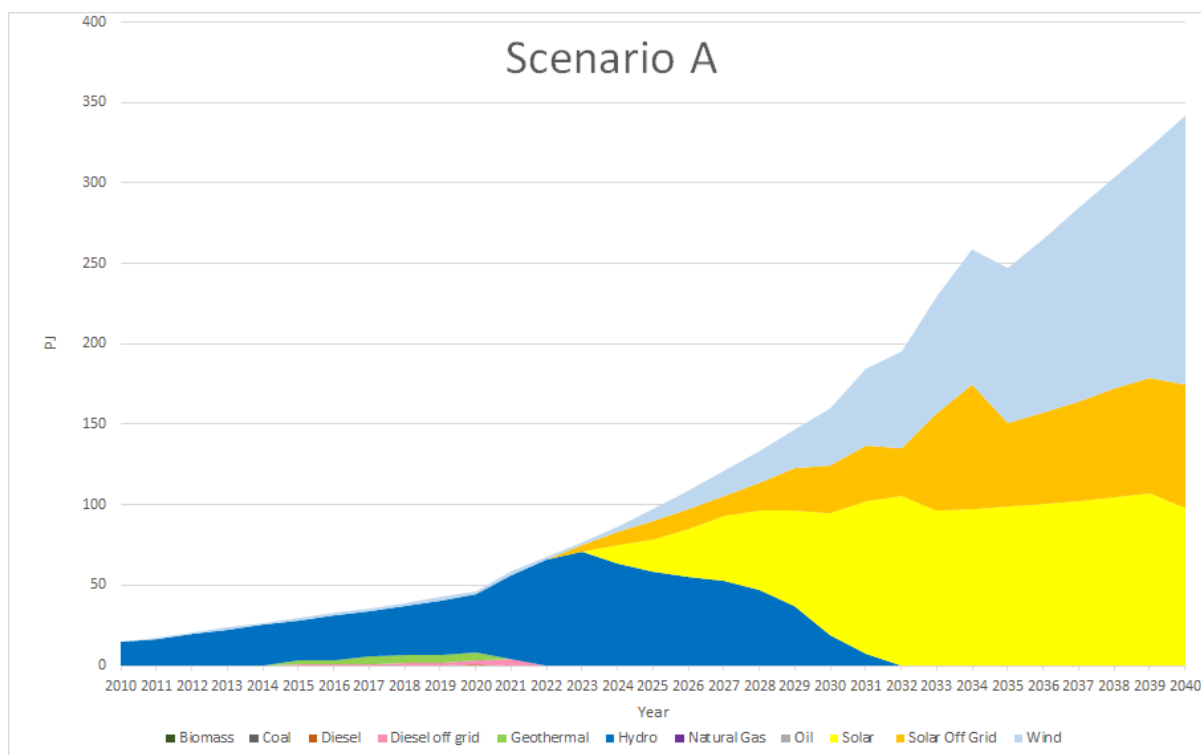


Figure 13: Total production by technology per year, scenario A

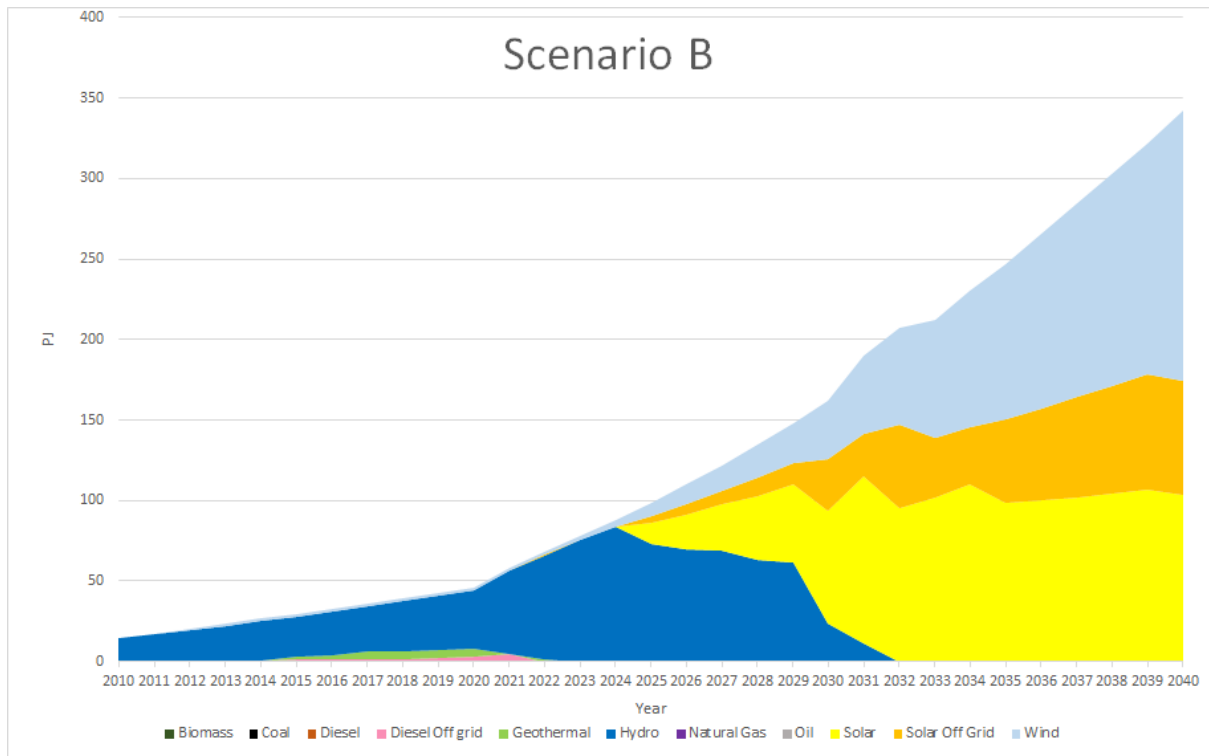


Figure 14: Total production by technology per year, scenario B

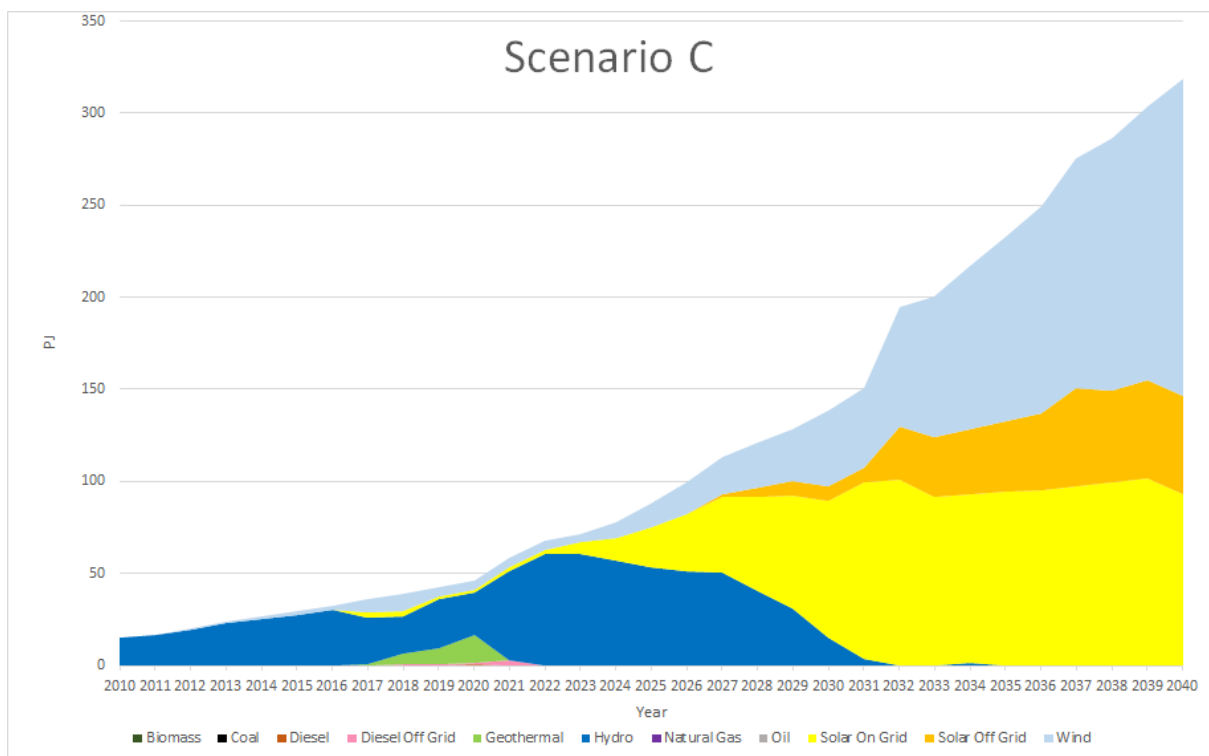


Figure 15: Total production by technology per year, scenario C

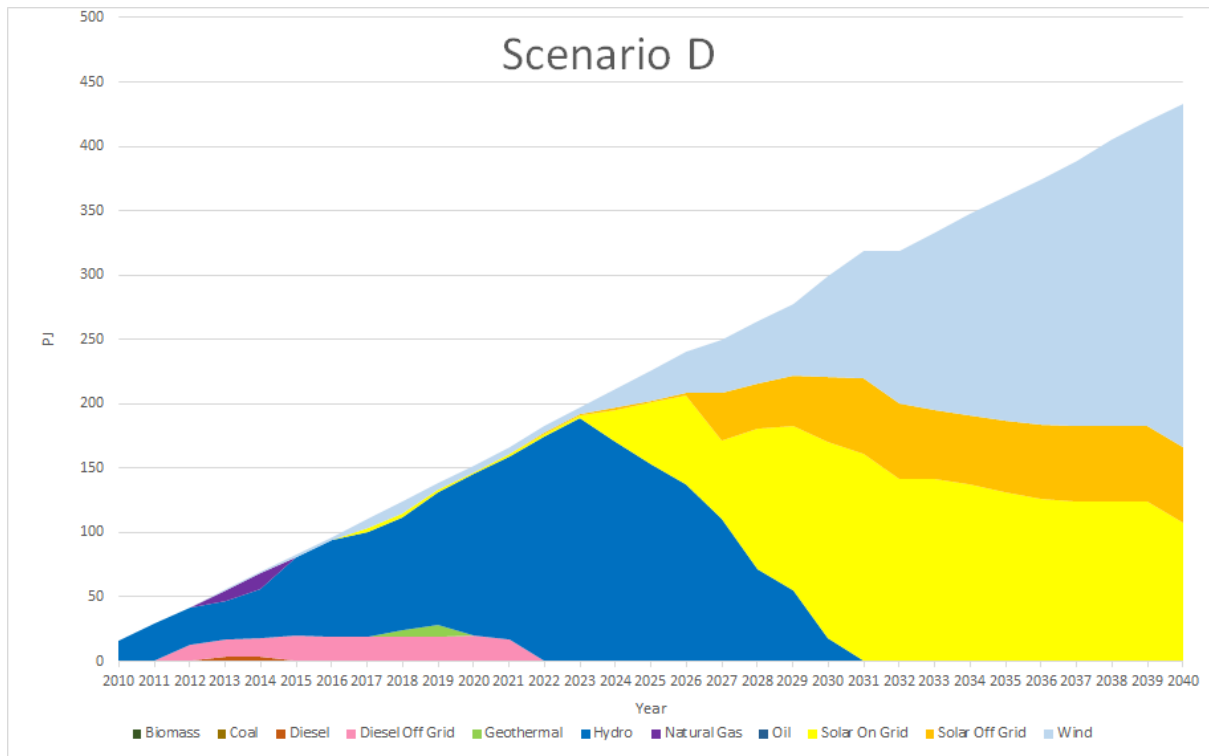


Figure 16: Total production by technology per year, scenario D

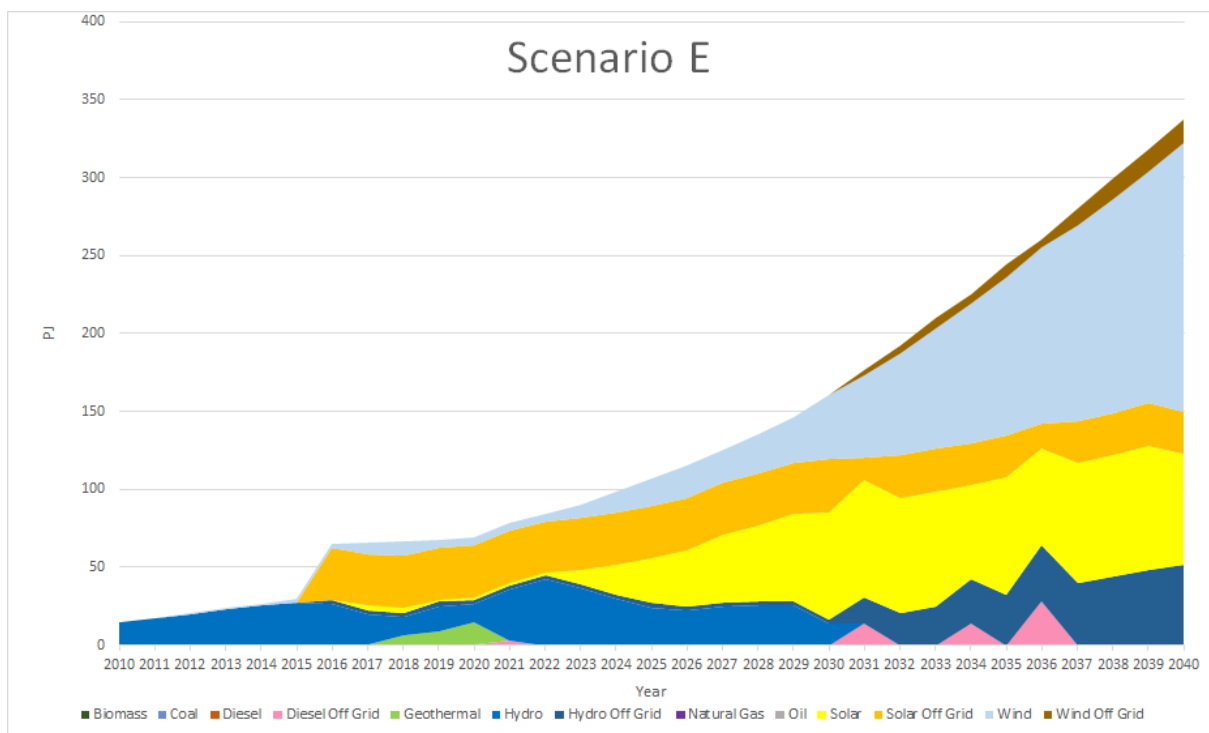


Figure 17: Total production by technology per year, scenario E

Table 2: Production by technology in 2030

Production in 2030 (PJ)					
Technology	A	B	C	D	E
Biomass	0	0	0	0	0
Coal	0	0	0	0	0
Diesel	0	0	0	0	0
Diesel Off Grid	-	-	-	-	0
Geothermal	0	0	0	0	0
Oil	0	0	0	0	0
Hydro	19	23,39	15,56	17,76	14,02
Hydro Off Grid	-	-	-	-	2,70
Natural gas	0	0	0	0	0
Solar	75,43	70	74,13	153,03	68,48
Solar Off Grid	29,95	32,09	29,95	50,34	34,30
Wind	35,80	36,52	40,55	78,30	41,11
Wind Off Grid	-	-	-	-	0,02
Total	160,19	162	160,19	299,44	160,63

4.4 Integration with OnSSET

Results obtained in OnSSET after the integration are presented below. Additional data from the integration can be found above, marked as scenario E.

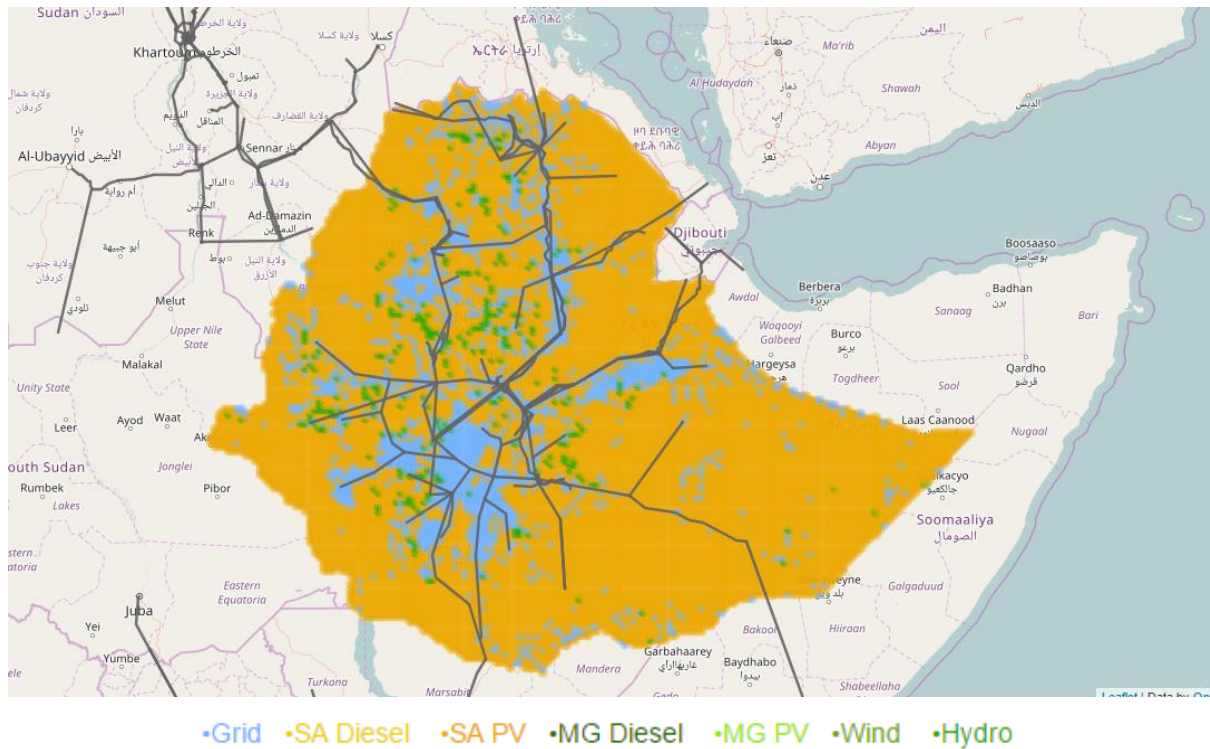


Figure 18: Map of technology split, scenario E

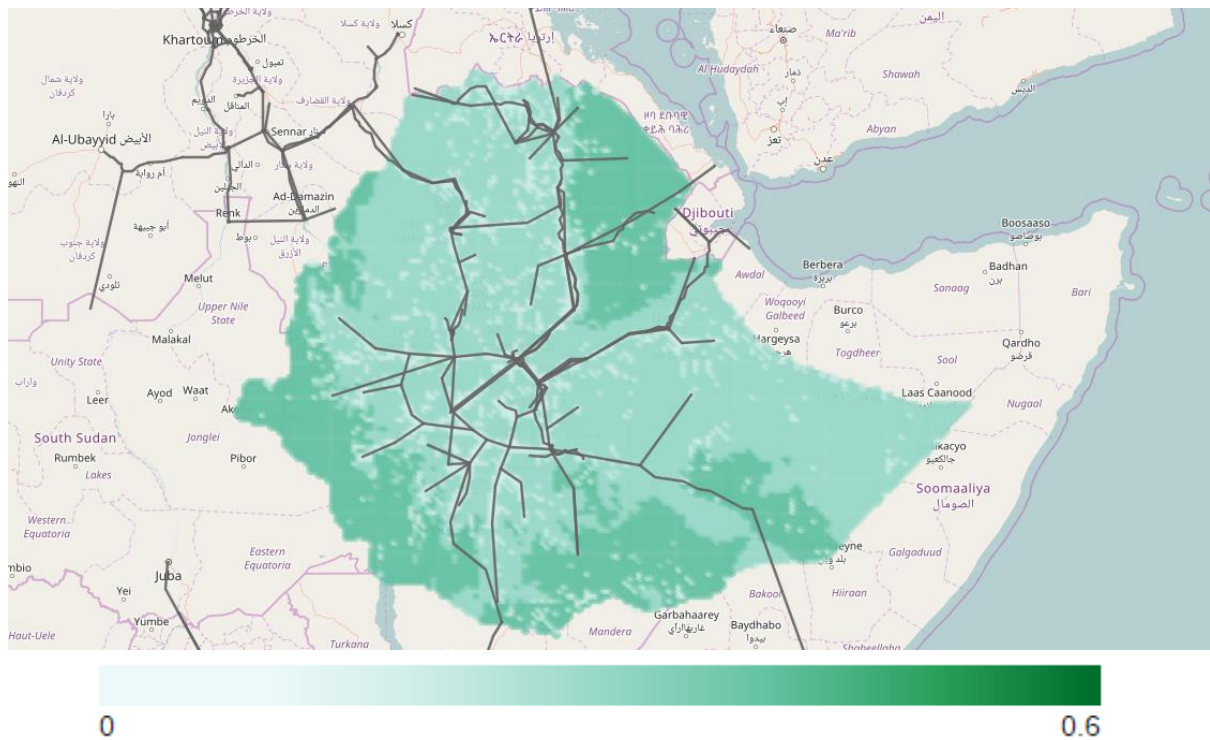


Figure 19: Map of electricity cost in USD/kWh, scenario E

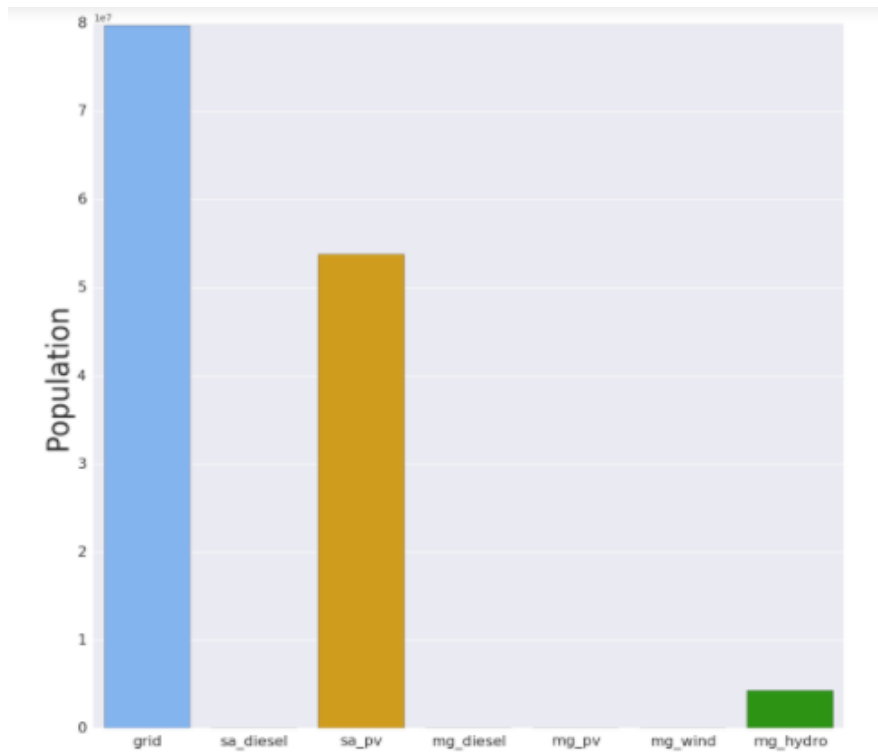


Figure 20: Technology split displaying the amount of people obtaining electricity via the different sources in 2030. The y-axis represents number of people in terms of 10^7

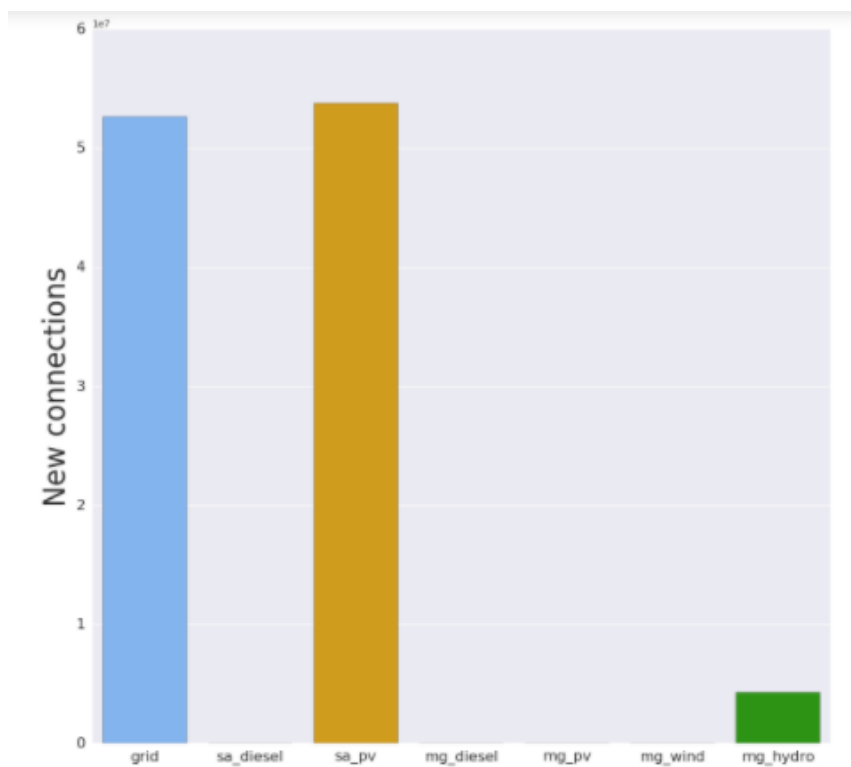


Figure 21: New connections obtained throughout the time period between 2015 and 2030, and the associated technology split. The y-axis is presented in terms of new connections multiplied by 10^7 .

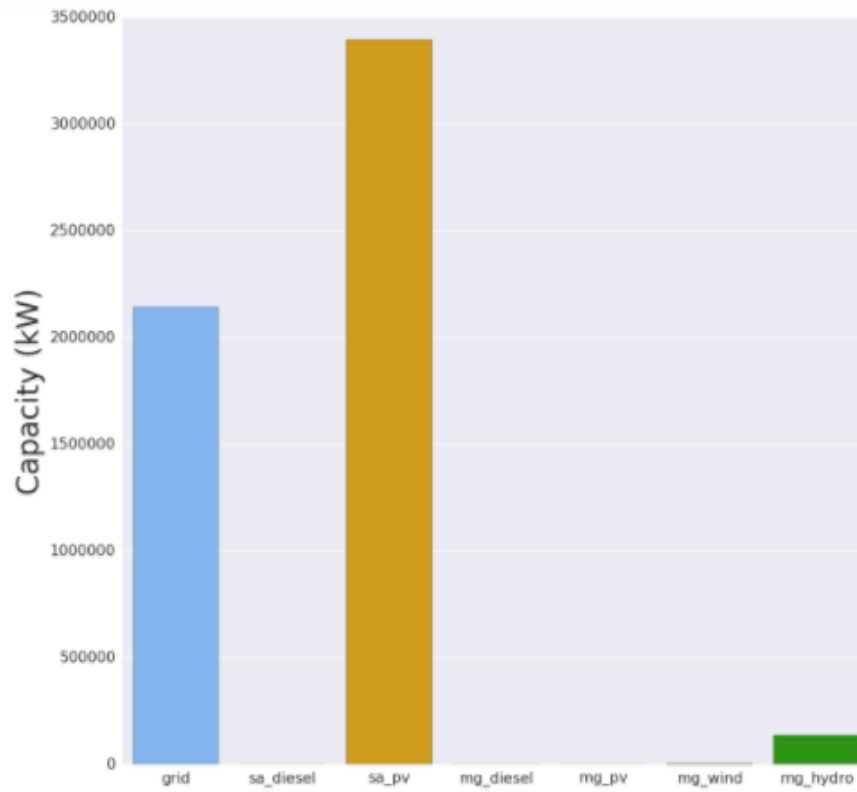


Figure 22: Capacity in 2030 and the associated technology split

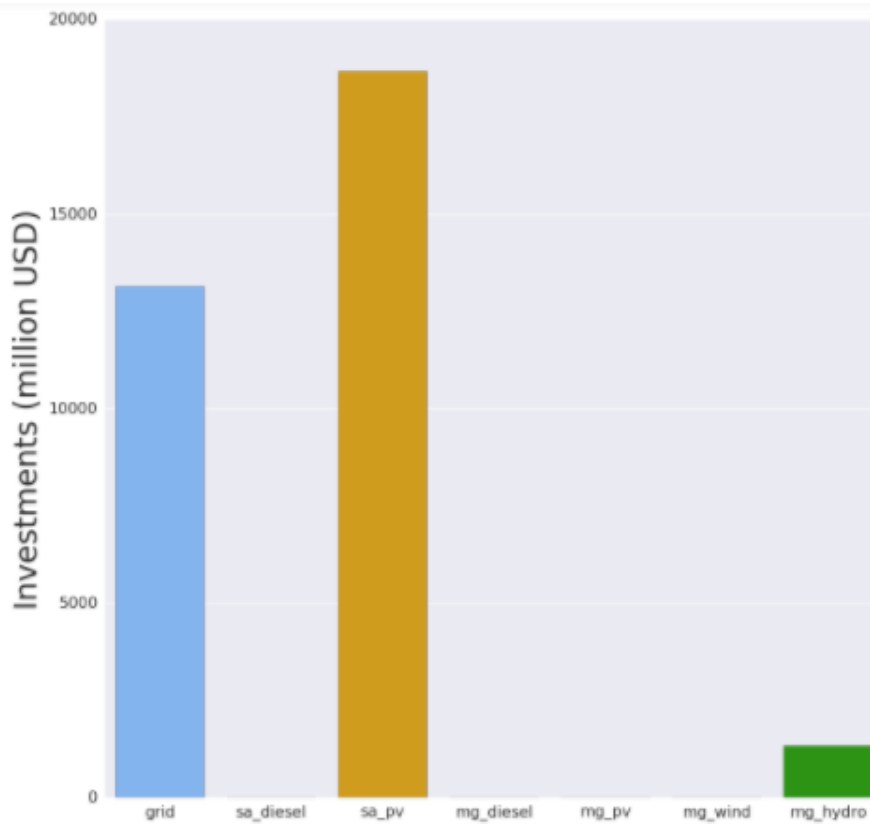


Figure 23: Investments required by each electricity source to meet the demand

5. Discussion

This chapter is a discussion over the obtained results.

5.1 Ethiopia and data-related uncertainties

This work is done based on energy modelling with the help from data about future plans and forecasts. It is thus interesting to discuss the difference between a theoretical model and the reality as well as the probability of reaching the goals. One factor contributing to uncertainties is the fact that it was hard to find historical data about the actual production taking place while it was relatively easy to find forecasts and future goals. It is quite common that arrangements do not turn out to be as planned and initiatives take longer time than intended. One example is The Reppie waste-to-Energy facility in Addis Ababa that was supposed to be producing energy by the beginning of 2017, as presented in the GTP II, but due to complications, for example regarding financial transfers, the constructions has been delayed and the manager can't say when the facility will be producing as planned.

The national goals are very ambitious, and Ethiopia has a great potential of having an energy system 100% based on renewable energy sources. The question is if the potential will be utilized given the financial- and time constraint.

Power outages are a big hindrance in Ethiopia's development as they influence the whole society and its prosperity. With no power, tons of devices stop working which affects private individuals as well as large businesses. For example, if a company produces perishables and its fridges shut down it loses loads of products which may lead to significant economic losses. Further, students often require a computer to, for example, search for information required for an assignment. If there are issues with the electricity stability, they will most likely also have trouble with charging of the computer or with accessing internet. This will in consequence slow down and make the whole studying process inefficient. If a hospital loses power it risks also damaging the functionality of medical equipment which may have direct consequences as severe as loss of human life. Power outages are also very discouraging for foreign investors which in its turn makes it harder for Ethiopia to establish itself on the international arena.

In order to avoid the many negative consequences of power outages, people are forced to rely on private diesel generators when the power supply from the national grid fails. This is not sustainable as it emits carbon dioxide and can be pretty expensive in the long run. It is thus also a question of independency, economy and environment. If Ethiopia wants to reach the SDG 7 they should directly start finding a solution to this problem and decrease the numbers of diesel generators around the country.

5.2 Annual Emissions

When looking at the charts of carbon emissions, figure 2-6, one can see that A and B are identical, and that C follows a similar pattern since it has its emissions during the same years but in a lower amount. The three first scenarios reach their maximum emissions in 2021 with a value of 1,5 ton in scenario A and B, and 1,1 ton in scenario C. Scenario D has the largest amount of emissions, and exceeds up to 5 tons in 2014 and 2021. Furthermore, one can observe an addition of a fuel - natural gas, for scenario D, in figure 5. This can be explained by the increased demand which may require a bigger production by current, established technologies in order to be satisfied.

Scenario E has a slightly different curve (see figure 6) and emits a lot of carbon dioxide from 2030 onwards. Since the scenario is created by an integration with OnSSET - a tool which in this case only runs up to 2030 - the results after that year are unreliable. The relevant results are within the modelling period, 2010-2030. During this period, the largest emissions occurs in 2021 and correspond to a value

of 0,85 ton. This means that scenario E is the most environmentally friendly scenario, if only considering emissions from 2010-2030.

When assuming scenario E only presents relevant results up to 2030, one can conclude that by 2030 none of the presented scenarios emit any carbon dioxide emissions, thereby all electricity is derived from renewable sources. The emissions in scenario A, B and C all sharply end in 2022. This is because from this year and onwards the model starts to produce electricity solely from renewable sources, see figures 13-16.

The time-span in which these relatively high carbon dioxide emissions occur is the nearest future. The phenomenon can be explained by the fact that some longer time is required in order to establish renewable energy sources and reach sustainable satisfaction of the demand. The demand is high but the renewables cannot produce enough, so alternative forms such as diesel and natural gas, are used.

A common desire that pertains to today's society is to become carbon free in order to decrease the negative effects GHG emissions have on the climate. The aim is to sustain the planet one is dependent on. However, social development and environmental protection are often in conflict as it is generally difficult to achieve one without disrupting the other. Countries such as Ethiopia face a challenge trying to achieve social stability by granting every inhabitant access to electricity, and at the same time becoming environmentally friendly through the exclusive usage of renewable energy sources. At times choices must be made and one has to prioritize - it might be worth it, to allow some emissions if that means that more people get access to electricity and therefore improve their life standard and other. One can further discuss the moral perspective by considering the fact that, in the past, the majority of global environmental destruction has been caused by the so called “western developed economies”. These economies achieved high living standards relatively early and led a lifestyle that required a lot of carbon dioxide emitting energy for many years, which resulted in great environmental damage, with noticeable traces still left today. One can ask themselves whether these actors should consequently take greater responsibility when it comes to environment protection, possibly supporting those who have not achieved an equally extensive development, and have thus not caused as much pollution et cetera.

5.3 Capital Investments

Capital investment was of particular interest when comparing scenario B with scenario A. The difference between the scenarios was that the residual capacity for distribution and transmission was degraded in the first one which presumably made the model invest more than usual in those areas. The increased total capital investment for scenario B was thus expected. In table 1 it is shown that the investments in the distribution and transmission system are the same for scenario A and C, since C was created from A and no changes were made regarding the distribution and transmission this was also expected.

The assumption, when making scenario B, was that the additional investments would lead to improvements in transmission and distribution. As mentioned in section “3.4.2 Scenario B”, flaws in distribution and transmission were, amongst other, contributors to the frequent power outages. By improving the distribution and transmission system one can get rid of the power outages and obtain a more sustainable energy system where the need of diesel generators is reduced.

Further, regarding the total capital investments, in table 1 and figure 7, one can observe that scenarios D and E involve the largest amount of investments. This can be explained by the significantly increased demand that was applied for these scenarios. It is worth mentioning that the increased demand also rises due to an increased export - an expenditure that can be turned into a valuable income for the country.

Scenario B is the least expensive and would be sufficient to meet the demand. It is however important to stress the fact that the scenarios are not comparable with each other as they vary greatly in input of among other the demand. One can argue that the most expensive case, scenario E, is the most complex

and realistic one as it includes the geospatial perspective. It takes into account the geographical conditions when determining whether a certain area should have its electricity supplied by the grid or other technologies, and in addition to that, it also presents the electricity cost as a changing variable depending on the distance to the grid and the used technology et cetera (see figure 18-23).

In order to enable the production of a new technology, investments need to be made for that technology. Therefore, figures 8-12, showing the investments per year for each scenario, represent the same kind of information as the production by technology, presented in the sections below.

5.4 Production by Technology Annual

In appendix I, information about relevant national policies used to create the scenarios are presented. National policies such as GTP II and the Master Plan aim towards an electrification rate of 90% by 2020 and 100% by 2037. Based on this trend one can assume that 100% of the population will have access to electricity by 2030 and thus SDG 7 will be achieved. In addition, the demand from TEMBA is based on a forecast from IEA and includes a growing demand from a growing population which most probably can be seen as representing the demand from the whole population by 2030. Scenario E was created to see how the population would get their electricity and what the costs would be to extend the grid or create mini grids, the demand was set so that everyone would get electricity by 2030 and that the average electricity consumption was 695 kWh/hh/year. One can thus argue that all scenarios will ensure electricity for all inhabitants by 2030.

In table 2 one can see that by 2030 all the scenarios have the same ranking between the existing technologies, with solar on grid being the most prominent one, followed by wind, solar off-grid and lastly hydro. Results from OnSSET present a similar trend, showing that among technologies not connected to the grid, solar is by far the most used energy source (see figures 20-23). Scenario E has a larger mix of different technologies and has the ranking as mentioned, followed by hydro- and wind off grid. Figures 13-16 show that scenarios A-D have a greatly increased penetration of solar power from around year 2022. At that time hydropower also starts to decrease, perhaps due to the fact that more cost-efficient alternatives are available. Scenario E has a production from solar off grid already from 2015 but follows a similar pattern as the A-D scenarios, with a large penetration of solar on grid from 2022 and onwards (see figure 17). One can also see an increasing amount of wind power during these years. This higher penetration of renewable sources might be explained by the fact that these technologies become more developed, more efficient and thus less expensive over the years.

Regarding scenario E, figure 18 shows in which geographical area each technology should be used for electricity supply and figure 19 presents how much each kWh would cost in different areas of the country. One can thus see that it is more expensive to supply electricity to those living far away from the national grid and that a big fraction of the population will get their electricity from stand-alone solar PV.

In the results for 2030 scenarios A-E show a production based on three different energy sources; hydropower, solar energy and wind energy (see table 2). One can find it interesting to discuss whether this energy mix is sufficiently diversified and constitutes a resilient electricity system. It is important to have many energy sources as that reduces the risk of paralysis of the society that can occur when you rely on one resource and that resource fails. On the other hand, other than hydro, solar and wind, there are not many renewables left to implement. Also, in the future there might be better options available for storage and therefore the reliability of the covered renewable sources might go up.

From the beginning of this project a big question regarding hydropower has been the conflicts with other aspects of the society it creates. Ethiopia is highly dependent on water resources as about 86% of the current electricity is being generated by hydropower and agriculture makes up a big part of the country's economy, having more than 80% of the population working in the sector. This, in combination with the fact that Ethiopia has a distinct dry season and is vulnerable to climate change leads naturally

to the question: is there enough water to supply all of the needs? Are there no clashes between energy and food, especially during the dry season?

Based on the national policies and reports such as GTP II one can assume the officials are not worried about the above-mentioned issue, as the statistics included in those documents state that only 5% of the total potential for hydro power is being used. The question remains amongst private people since Ethiopia suffered from a drought in 2016, resulting in more than 18 million people in the need of food aid.

Another fact contributing to the uncertainty about hydropower is that the water running through Ethiopia also is a valuable resource for neighbouring countries, for example Egypt. As it is a shared resource Ethiopia's utilization of it might affect Egypt negatively which in turn might create tension between the countries. Hydropower is therefore not only a question of energy, but also health, economy and politics.

6. Conclusions

The objective of the study was to find a configuration of the energy system that would benefit the economy, the society and the environment. By using two modelling tools that run a least cost optimization all presented scenarios are cost efficient based on their given constraints and would therefore benefit the economy. The climate and environmental aspect is a bit more complex but was significantly simplified in this project by delimiting greenhouse gases emissions to carbon dioxide. In an environmental perspective the results are positive since all energy is produced by renewables and the emissions are non-existent by 2030. The benefit of the society is straightforward in a sense that a population gets a higher quality of life and a greater possibility to development in other sectors, as for example education and health, with an increased electricity access.

The scenarios have all been given different constraints regarding, among other, the demand which makes a comparison difficult, if not impossible. It is not just to compare scenario B with scenario E in terms of capital costs for example, as scenario E has a much greater demand input. Nevertheless, some common patterns are noticed for all scenarios; with time, solar and wind become the most used technologies and all carbon dioxide emissions cease to exist by 2030. Though this one can conclude that all scenarios will lead to an achievement of SDG 7 - sustainable energy for all.

7. Suggestions for future work

Throughout this project questions have arisen that would widen the extent of the project and bring out more accurate results. Some of them are presented below as suggestions for future work in order to make the model and the results more realistic.

- Take into account the energy losses that occur in the transmission and distribution system
- Include other forms of energy, not only electricity
- Include trade with neighbouring countries
- Examine the amount of other emissions from each scenario that contribute to climate change
- Finally, modelling conducted in this project is partially based on Ethiopia's GTP II. It would be interesting to study a follow-up to the plan and see whether the estimations and goals presented were reached.

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Appendices

Appendix I - Country data

Electrification rate

Table 3: Percentage of population without access to electricity

Amount	Year	Source
73%	2014	(IEA, 2014)
40%	2015	(National Planning Commission, 2016)
45%	2015	(Asnake & EEP, 2015)
75%	2016	(Climatescope, 2016)
73%	2016	(Energypedia, 2017)
70%	2016	(Girma, 2017)

As seen in the table above different sources present different values for the amount of people without access to electricity. In the report, the number from IEA is used. The reason behind this is partially because IEA is a reliable and impartial source but also because three other sources presents similar numbers for 2016. The numbers from 2015 are both from Ethiopian sources. One explanation to the fact that these numbers are lower might be that the country aspires to display a better image to the global community in terms of development, than it actually is. There is also a possibility that Ethiopian organisations have access to data that is not available for international organisations. It might also be a question of definition. Some may define having access to electricity as being x km from the grid, and others might argue that one has electricity when there is a direct connection and the considered individual can actually use the electricity.

Relevant national policies

Growth and Transformation Plan II, GTP II (National Planning Commission, 2016)

The GTP II is a five-year plan (2015/16-2019/20) regarding economic and social development, built on sectoral policies, strategies and programs. The major objective is to serve as guidelines towards realizing Ethiopia's vision of becoming a middle-income country by 2025. Regarding energy infrastructure the goal is to provide 90% of the population with electricity by 2020. Improvements have been made upon lessons from the implementation of GTP I (2010/11-2014/15) as well as the post-2015

SDGs. It has been created as a collaboration between high level political leaders and relevant stakeholders at regional and federal levels. Before publishing the first version, approval from the council of ministers and parliament was obtained (National Planning Commission, 2016).

Climate-Resilient Green Economy, CRGE (UNDP, 2011)

The Climate-Resilient Green Economy initiative (CRGE) was released in 2011. It was created by the Prime Minister's office, the Ethiopian Development Research Institute (EDRI) and the environment Protection Authority (EPA). The strategy explains a sustainable and climate-resilient pathway for Ethiopia towards a low carbon economic growth. The vision is to become a middle-income country by 2025 and the development is based on four pillars; transport, power, agriculture and forestry. Important initiatives regarding the power sector are to produce renewable electricity, majority from hydro, and export to the Eastern African countries. The government aims at expanding the electricity supply by 14% per year in order to support economic growth (UNDP, 2011).

Ethiopia's Intended Nationally Determined Contribution, EINDC (Federal Democratic Republic of Ethiopia, 2016)

At the United Nations Climate Change Conference in Paris 2015 Ethiopia signed the United Nations Framework Convention on Climate Change (UNFCCC, 2015). Thereafter the officials published the Ethiopian INDC (EINDC) which presents how Ethiopia can contribute to the fight against climate change and simultaneously improve the welfare of its citizens. The target year is 2030 and the plan includes near-, medium- and long-term actions on how to reduce GHG emissions from agriculture, forestry, energy, transport, industry and buildings. The means of implementations are presented as well as the method of monitoring and evaluation (Federal Democratic Republic of Ethiopia, 2016).

Ethiopian Power System Expansion, Master Plan Study (Brickerhoff, 2014)

The first version of the Master Plan was released in 2003 followed by an updated version in 2005 and was then published in September 2014 after being reviewed by Parsons Brickerhoff. The study was conducted in collaboration with Ethiopian Electric Power, EEP, the Ministry of Water and Energy along with other government organisations. It contains forecasts on load and distribution, planning on generation and distribution and a report on financial assessment and tariff impact. The objectives are to find the least cost expansion program for Ethiopia's generation and transmission system from 2012 up to 2037. The forecasts were based on a 100% electrification rate by 2037 (Brickerhoff, 2014).

Appendix II - Creating scenarios

In the following section all input-data used when creating the scenarios are presented.

Scenario A

Modeling, step by step

In order to get a distribution and transmission system by the base year, the following steps were executed:

1. Created a scenario within the model *Ethiopia*. Named the scenario *Ethiopia_A*
2. Looked at the results for the AccumulatedNewCapacity for the "Distribution technology tertiary to industrial electricity demand". The value for the base year, 2010 was:
0,468436040446633
The other distribution systems, for Rural and Urban were zero in 2010, due to the fact that the

modelling hasn't started yet)

3. The value, obtained in step 3, was taken as an input for the ResidualCapacity for the distribution system, called ETEL00TDI. The value was kept constant and entered as the same value for the years 2010-2040. Initially this value was 0 for all years, indicating that there has been no distribution before the modelling started. Now, with a capacity factor for the distribution the first year it means that there is a distribution system at the base year.
4. The same process was done for the transmission system
5. Looked at the results, for the model without the ResidualCapacity , AccumulatedNewCapacity for the “Transmission technology secondary to tertiary”. The value for the base year, 2010 was: 0,484772886729414
6. The value, obtained in step 6, was taken as an input for the ResidualCapacity for the transmission system, called ETEL00T00. The value was kept constant and entered as the same value for the years 2010-2040.

Scenario B

Modeling, step by step

In order to get a distribution and transmission system that improves with time a gradually decreasing residual capacity is entered. This forces the model to invest more money into the distribution and transmission system which leads to improved systems.

1. Cloned the scenario *Ethiopia_A*, named it to *Ethiopia_B*
2. Looked at the results for the AccumulatedNewCapacity for the “Distribution technology tertiary to industrial electricity demand” in scenario A. This value was 0,468436040446633 in 2010. The reason to why the value is taken from scenario A instead of B is that the value in the new scenario is zero the first year, since the residual capacity already is enough, so there is no need for an accumulated new capacity.
3. Also looked at the results for the AccumulatedNewCapacity for the “Transmission technology secondary to tertiary” in scenario A. This value was 0,484772886729414 in 2010.
4. Took the value from step 2 and 3 and put it in Excel in order to be able to gradually decrease the value over year. The rate was chosen to be 10% per year. Instead of seeing the improvement as investments in the old systems in order to make it better it can be seen as replacements of the old systems. With this approach, a decreasing rate of 10% per year implies that it will take 10 years to replace the distribution system.
5. With a rate of 10% the values for the distribution system stretch from 2010 to 2019 and the ones for the transmission stretch from 2010 to 2020. These values were copied from excel and

entered as input for scenario B for ResidualCapacity. The values for the following years, from 2019 and 2020, up to 2040 were set as zero.

Scenario C

Compilation of data

For the creation of scenario C, data regarding various technology capacities was collected from the GTP II and Ethiopia's Master Plan for the electricity sector. If contradicting information was given by the two of them, GTP II was used as it is a more recent, and thus updated document. The Master Plan was used for data which was not found in GTP II. The GTP II is limited to the years 2015-2020 while the Master Plan presents numbers from 2010 to 2019, so for the first 5 years of the modelling, the second one was used unless it contained values clearly dismissed by the GTP II.

For facilities that have adapted a constant capacity value for more than the two last years of the presented years, the same constant value was applied for the rest of the years until 2040, unless it was said to be retired before. For example, if facility A was presented by the Master Plan to have a constant capacity of 50 MW for 2017, 2018, 2019 it was assumed it would continue to have the same capacity each year until the end of the modelling period or retirement, depending on which came first. It was assumed that no degradation took place.

Facilities were then divided into those existing before the base year and those created during or after the base year. Those already established by 2010 had their capacity put into ResidualCapacity. Power plants constructed after 2010 had their capacity summarized and if any difference occurred between continuous years, it was taken as additional investment in capacity and put into TotalAnnualMinCapacityInvestment. For 2010-2020 the same data was also put as TotalAnnualMaxCapacityInvestment in order to force the model to operate exactly accordingly to the given policies. Other years the model was free to invest more in the various technologies.

For technologies where, for the same year, larger capacities were presented by the GTP II than the Master Plan, the difference was assumed to be new investments and added into NewCapacityInvestments. This assumption was based on the fact that the GTP II was published one year later than the Master Plan and new projects have likely been initiated during that time period. If any power plant was retired before 2040 the extrapolated capacities were adjusted accordingly.

If the capacity was larger in the Master Plan than in the GTP II, the difference was neglected and only the value given by GTP II was taken into account. This is because a smaller capacity in GTP II indicates that some investments that were predicted in the Master Plan did not occur and were thus not considered in the updated policy.

For example, in case of sugar, all the plants were created after 2010 and thus no residual capacity existed. The GTP II covered thus all the new investments and was used as the only data source for the years 2015-2040. The Master Plan was used between 2010-2014 and also gave information about retirement of two plants. The capacity of the two plants was 26 MW and was subtracted from the extrapolated value of 474 MW for the last 2 years, when the plants were no longer operating. If any difference in capacity occurred between consecutive years it was taken as investments in new capacity. For 2010-2020 the same value was set for both TotalAnnualMinCapacityInvestment and TotalAnnualMaxCapacityInvestment, and for 2020-2040 only TotalAnnualMinCapacityInvestment was set.

Sugar, waste and biomass were added together to form an input to the Biomass CHP plant (ETBMCHP00). This was done due to the fact that there are no separate technologies for waste and sugar in MoManI and they can all be seen as one.

For hydropower, it was determined which facilities are large (Dam hydro power plant, ETHYDMS02, in MoManI) and which are small (Small hydro power plant, ETHYDMS01). According to the U. S. Department of Energy (U. S. Department of Energy, n.d.) large hydropower plants have a capacity of 30 MW or more, whilst small ones have a capacity of 10 MW or less. This information was used to determine the type of the various plants. In this project, Tis Abay I was categorized as a small hydro power plant even though its capacity is 11 MW. This was done due to the fact that 11 MW is much closer to the capacity of a small hydropower facility than that of a large one.

Additionally, information on hydropower plants was obtained from journal articles written by amongst other Q. Goor (Goor, Halleux, Mohamed, & Tilmant, 2010) and Ruben Muller (Müller, Gebretsadik, & Schütze, 2016). These sources determined which facilities are run-of-river (ETHYDMS03).

Furthermore, any differences between the GTP II and the Master Plan were taken as investments into Dam hydro power plants. That is because it's the largest type of hydro facility and most investments are put into it in Ethiopia (National Planning Commission, 2016).

Solar power was omitted in the Master Plan and only information about solar power facilities connected to the national grid was included in the GTP II. Data about solar off-grid was difficult to find, other than articles (The World Bank, 2016) about smaller project where private investors distributed solar lanterns or Solar Home System (SHS) to people in the rural areas of the country no reliable information was obtained. In comparison with the capacity on the national grid, these numbers of smaller solar systems were negligible. The input data to the model was therefore chosen to only consider the given numbers from the GTP II, which is just input to the national grid. Due to lack of more detailed information the total solar capacity was divided equally between the four technologies connected to the national grid (ETSOC1P00, ETSOC2P00, ETSOC3P00, ETSOU1P03). Similar division occurs for wind, where the total capacity was divided equally between Wind onshore 20% CF (ETWI25P00) and Wind onshore 30% CF (ETWI30P00).

Diesel is often used as an off-grid technology or as a back-up when there is no supply from the national grid in Ethiopia. Due to the fact that other technologies have a promising potential of producing large amounts of clean energy to the national grid, diesel was only considered as an off-grid technology and distributed equally between the three diesel plants that are directly connected to the demand (ETDSRCF12, ETDSRCFU1, ETDSRCP03).

As focus is directed towards alternative renewable sources in the energy planning in Ethiopia, the GTP II and Master Plan assume none or very little generation from coal and oil. A possibility for future usage and exploitation is however considered if the conditions are fitting. This is reflected in the model by setting TotalAnnualMinCapacityInvestment for coal and oil to 0. This allows for investments if necessary and does not constrain the technology completely.

Furthermore, no data was found for natural gas, thus the minimum capacity was set to 0, allowing the model to invest whatever amount it calculates as optimal.

The demand was obtained from TEMBA data.

Modeling, step by step

1. Cloned *Ethiopia_A*, named it to *Ethiopia_C*
2. Started off with wind. Entered ResidualCapacity for ETHYDMS02, ETHYDMS03 and ETHYDMS01. Checked the TotalAnnualMaxCapacity for the same technologies (ETHYDMS02, ETHYDMS03 and ETHYDMS01). The values for ETHYDMS02 were lower than the ResidualCapacity so they were changed to default value (99 999). This was done because the total max capacity can't be lower than the residual capacity.
3. Entered the TotalAnnualMinCapacityInvestment for ETHYDMS02, ETHYDMS03 and ETHYDMS01. Changed the TotalAnnualMaxCapacityInvestment for ETHYDMS02, ETHYDMS03 and ETHYDMS01 to default value (99 999).
4. Changed the TotalAnnualMaxCapacity for ETHYDMS01 to default value (99 999) since the values was lower than the TotalAnnualMinCapacityInvestment.
5. Moved on to wind. Entered TotalAnnualMinCapacityInvestment for ETWI25P00 and ETWI30P00. Checked the TotalAnnualMaxCapacityInvestment for ETWI25P00 and ETWI30P00. Changed them to default value (99 999)
6. Moved on to Geothermal. Entered the ResidualCapacity for ETGOCVP00. Entered TotalAnnualMinCapacityInvestment for ETGOCVP00 and changed the TotalAnnualMaxCapacityInvestment to default value (99 999)
7. Moved on to Biomass. Entered TotalAnnualMinCapacityInvestment for ETBMCHP00 and changed the TotalAnnualMaxCapacityInvestment to default value (99 999)
8. Moved on to Diesel. Entered TotalAnnualMinCapacityInvestment for ETDSRCF12, ETDSRCFR1 and ETDSRCFU1.
9. Moved on to Solar. Entered TotalAnnualMinCapacityInvestment for ETSOC1P00, ETSOC2P00, ETSOC3P00 and ETSOU1P03. Checked the TotalAnnualMaxCapacityInvestment for ETSOC1P00, ETSOC2P00, ETSOC3P00 and ETSOU1P03 changed them to default value (99 999)
10. In order to force the model to follow the governmental plans the same values for TotalAnnualMinCapacityInvestment were entered for TotalAnnualMaxCapacityInvestment for the years 2010-2020. The TotalAnnualMaxCapacityInvestment was entered for the following technologies: ETHYDMS02, ETHYDMS03, ETHYDMS01, ETWI25P00, ETWI30P00, ETGOCVP00, ETBMCHP00, ETDSRCF12, ETDSRCFR1, ETDSRCFU1, ETSOC1P00, ETSOC2P00, ETSOC3P00 and ETSOU1P03. For the same technologies, the TEMBA data was entered between 2020 and 2040 to set a reasonable limit for investments.

Scenario D

Compilation of data

Scenario D was created to show the results of a higher demand. In the CRGE it is presented that the electricity production can in the future exceed the national demand, due to improved energy efficiency, and therefore open up for possibilities to export to neighbouring countries.

When calculating the demand, data from the CRGE strategy was used. Two values were given - in 2010 the demand was 4 TWh and for 2030 it was forecasted to be 75 TWh - and a linear extrapolation was conducted for the other years. In order to divide the total into the I-, U- and R-sector TEMBA data was studied and the same percent division was applied. For example, if I constituted 60% of the total demand in TEMBA-data it was also assigned 60% of the total demand from the CRGE-data in the new scenario. This was done due to the fact that no other data was found regarding the division between the three sectors.

In order for the model to be able to invest more so that the increased demand could be met the TotalAnnualMaxCapacityInvestment was changed. To find a reasonable escalation a percent was obtained by dividing the new demand from the CRGE strategy with TEMBA-based demand for each year. This factor was then multiplied with the TotalAnnualMaxCapacityInvestment. In scenario C, which was the template for this scenario, many of the technologies had the capacity investment set to 0 in the first years. These hindering values (0) between 2010 and 2020 were adjusted to 1.

Modeling, step by step

1. Cloned *Ethiopia_C*, named it *Ethiopia_D*
2. Entered the SpecifiedAnnualDemand for I, R and U for 2010-2040 (based on the CRGE data)
3. Adjusted the TotalAnnualMaxCapacityInvestment. First it was a version from scenario C. This maximum capacity was adapted to the increased demand. A percent was obtained by dividing the new demand from the CRGE strategy with TEMBA-based demand. TotalAnnualMaxCapacityInvestment was then multiplied by the obtained factors.
4. Additionally, to open up for more investments and to meet the increased demand during the initial years all 0 were replaced by 1 from 2010 to 2020.

Scenario E

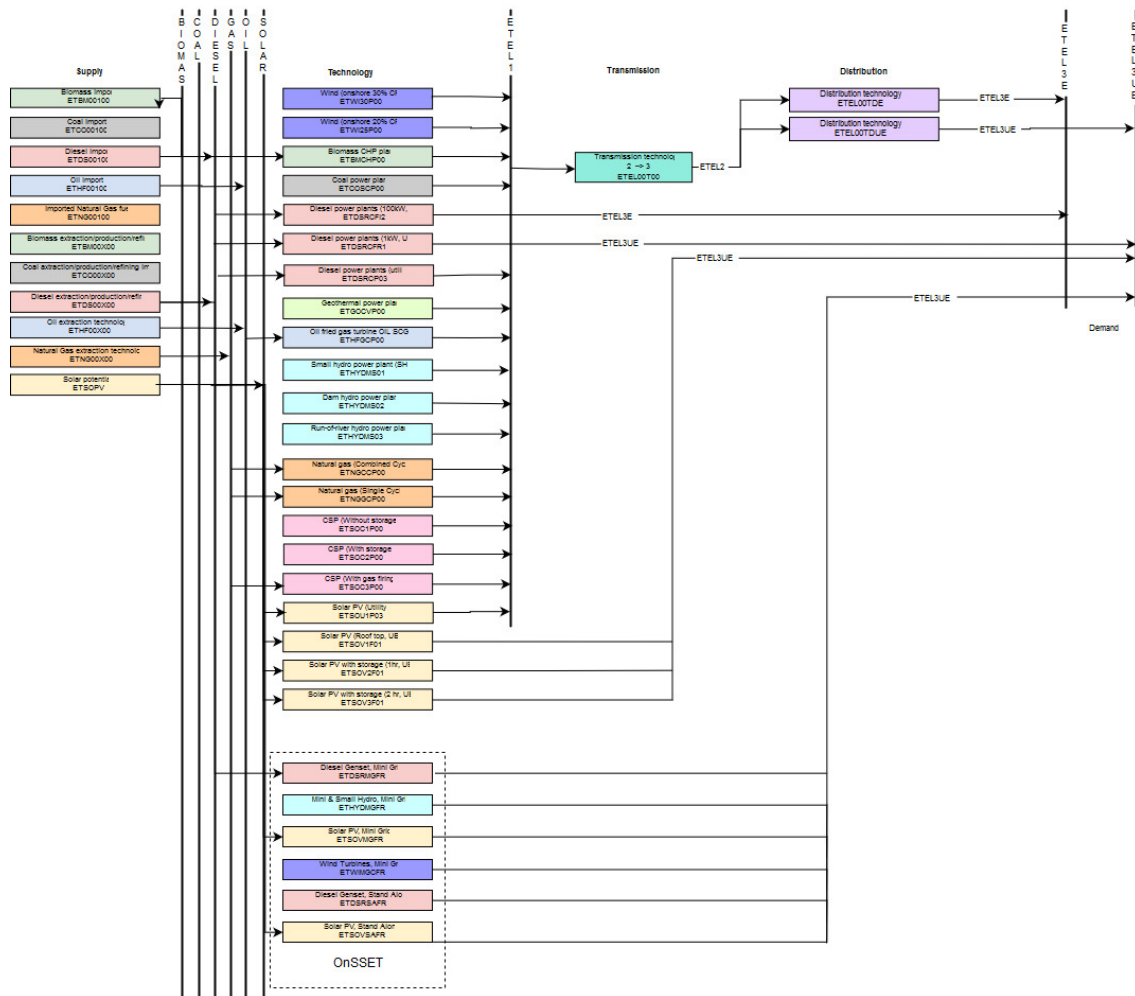


Figure x: Reference Energy System II, scenario E

Adjustment of OnSSET

In step 3b, the parameter called “scenario” which explains the demand in kWh/household/year in 2030, was set to 696. This number was chosen since the TEMBA data assumes an energy intensity for of 696 kWh in 203 for both rural and urban households.

In step 3c, the code was changed in order to get the right electrification rate, 0,27. This was done by setting the following numbers; min_night_lights = 33, pop_cutoff = 1700, max_grid_dist = 2,5, max_road_dist = 1,8. The code was set to the following;

```
df[SET_ELEC_CURRENT] = df.apply(lambda row: 1
    if row[SET_NIGHT_LIGHTS] > min_night_lights or
    (row[SET_POP_CALIB] > pop_cutoff and
    (row[SET_GRID_DIST_CURRENT] < max_grid_dist or
    row[SET_ROAD_DIST] < max_road_dist)) else 0, axis=1)
```

In step 3d, the diesel price was changed to be the same as the one in OSeMOSYS for 2030, which is represented as VariableCost. This was done by taking the value for ETDS00100 and ETDS00X00 for 2030, and calculating their mean value and converting it from M\$/PJ to \$/l. The obtained value, 0.917992 \$/l was then entered in OnSSET.

Adjustment of OSeMOSYS

The following changes were done in order to harmonise the two models with each other. For specific input numbers, see “appendix III - Modeling input”.

1. Cloned the original model *Ethiopia_2* and one of its scenarios, *scenario C*
2. Went to edit set data, technologies. Entered 6 new technologies.
All following changes were done for the 6 new technologies, from 2010-2040 in MoManI:
3. Went to Scenarios, entered CapitalCost, OperationalLife, CapacityFactor, AvailabilityFactor, FixedCosts, VariableCosts and InputActivityRatio
4. Entered CapacityToActivityUnit as 31,536 ·
5. Entered OutputActivityRatio as value 1 and linked to the residential demand (ETEL3R)
6. Entered ResidualCapacity as 0
7. Entered TotalAnnualMaxCapacity for all new technologies as default value (99 999 GW)
8. Entered TotalAnnualMaxCapacityInvestment as 0,25 GW from 2015 until 2040
9. New distribution lines were created to harmonise OSeMOSYS with OnSSET. The original U and R were added together in a new line, called UE, un-electrified, which represent the residential demand considered by OnSSET. This is due to the fact that OnSSET only takes into account the residential demand from households that aren't electrified in 2015 (modeling start year for OnSSET). The rest of the national demand was represented in a line called E, electrified, which is given by the total demand minus the un electrified residential demand.
10. In order to harmonise the demand from the two models the un electrified residential demand in 2030 was calculated by using the inputs from OnSSET.

Data from OnSSET:

kWh/household/year in 2030: 696

Number of people per household: 5

Population in 2030: 138 296 890

Calculated demand in 2030:

$$\rightarrow \text{UE-demand} = (138\,296\,890/5)*696$$

11. Since the start year in OnSSET is 2015, all demand before that year was covered by the electrified ones, E. For the years 2010-2015 the E-demand was set to the total demand given by TEMBA and UE-demand was set to zero. By dividing the calculated UE-demand in 2030 by 15 years (2030-2015) a smooth growth rate was found, which was then used to linearly extrapolate the UE-demand so it covers the modelling period (2010-2040). The E-demand was given by the difference between the total demand given by TEMBA minus the calculated UE-demand.

Integration

After running the OSeMOSYS model results were obtained and the Grid Cost could be calculated to then be used as an input in OnSSET. To calculate the Grid Cost, the following steps were conducted, all calculations were made in Excel:

1. Made an assumption that the economic life time can be set as the same as the operational life, since it is the period over which an asset expects to be usable.
2. Entered the discount rate to 0,05
3. The capital recovery factor, CRF, was calculated with the use of formula 1 and values from the economic life time for the technologies supplying the grid and the discount rate.

$$\text{Capital Recovery Factor} = CRF = \frac{r(1+r)^N}{(1+r)^N - 1} \quad (1)$$

$r = \text{Discount Rate}$

$N = \text{Economic Life of Technology}$

4. Extracted CapitalInvestment, DiscountedOperatingCost and ProductionByTechnologyAnnual for the technologies producing electricity to the grid from the results in MoManI and entered them into the Excel-sheet
5. Proceeded to calculate the Annual Investment Cost according to the following formula 2

$$\text{Annual Investment Cost} = AIC = \sum_{t=1}^n \text{CapitalInvestment}_t \times CRF_t \quad (2)$$

6. The Grid Cost per year was then obtained with the usage of formula 3

$$\text{Grid Cost per year} = \frac{\sum_{t=1}^n \left\{ \frac{\text{Annualised Investment Cost}_t}{(1+r)^t} + \text{Discounted Operating Cost}_t \right\}}{\sum_{t=1}^n \frac{(\text{Annual production } E_{TEL1})}{(1+r)^t}} \quad (3)$$

7. The grid cost obtained above gives a value for each year. In order to get one single input for OnSSET, a weighted average of these numbers, from 2010-2030, were obtained.

8. Entered the Grid Cost 0.12972181 in OnSSET and ran the model

9. Extracted the optimal electrification split given by OnSSET and calculated an upper and lower limit of the activity limit and entered it for the off-grid technologies in OSeMOSYS for the years 2015-2030. The parameters are called TotalTechnologyAnnualActivityLowerLimit and TotalTechnologyAnnualActivityUpperLimit in OSeMOSYS

10. Entered the SpecifiedAnnualDemand for UE given indirectly from OnSSET, by calculations in Excel

11. Ran MoManI and calculated a new Grid Cost which was entered in OnSSET. This iteration was done until the newly obtained Grid Cost was less than 10% different from the previously one

Appendix III - Modeling input

In the following section, all input-data used when creating the scenarios are presented.

Table 4: Naming of technologies

Name	Description
ETBACKSTOP	Backstop technology
ETWI30P00	Wind (Onshore, 30% CF)
ETWI25P00	Wind (Onshore, 20% CF)
ETBMCHP00	Biomass CHP plant
ETCOSCP00	Coal power plant
ETDSRCFI2	Diesel power plants (100kW, I)

ETDSRCFR1	Diesel power plants (1kW, R)
ETDSRCFU1	Diesel power plants (1kW, U)
ETDSRCP03	Diesel power plants (Utility)
ETGOCVP00	Geothermal power plant
ETHFGCP00	Oil fired gas turbine OIL SCGT
ETHYDMS01	Small hydro power plant (SHP)
ETHYDMS02	Dam hydro power plant
ETHYDMS03	Run-of-river hydro power plant
ETNGCCP00	Natural gas (Combined Cycle)
ETNGGCP00	Natural gas (Single Cycle)
ETSOC1P00	CSP (Without storage)
ETSOC2P00	CSP (With storage)
ETSOC3P00	CSP (With gas firing)
ETSOU1P03	Solar PV (Utility)
ETSOV1F01	Solar PV (Roof top, R)
ETSOV2F01	Solar PV with storage (1 hr, R)
ETSOV3F01	Solar PV with storage (2 hr, R)
ETSOV4F01	Solar PV (Roof top, U)
ETSOV5F01	Solar PV with storage (1 hr, U)
ETSOV6F01	Solar PV with storage (2 hr, U)
ETEL00T00	Transmission technology (Secondary to

	Tertiary)
ETEL00TDI	Distribution technology (Tertiary to I electricity demand)
ETEL00TDR	Distribution technology (Tertiary to R electricity demand)
ETEL00TDU	Distribution technology (Tertiary to U electricity demand)

Table 5: Time slices of a year in Ethiopia

Timeslice	Share of energy and time
Summer Day (SD)	0,25
Summer Night (SN)	0,25
Winter Day (WD)	0,25
Winter Night (WN)	0,25

Table 6: Default values

Parameter	Value	Unit
AccumulatedAnnualDemand	0	PJ
AnnualEmissionLimit	999 999	ton
AnnualExogenousEmission	0	ton
AvailabilityFactor	1	Ratio
CapacityFactor	1	Ratio
CapacityOfOneTechnologyUnit	0	GW

CapacityToActivityUnit	1	PJ/GW
CapitalCost	9 999	M\$/GW
CapitalCostStorage	0	-
ConversionId	1	-
ConversionIh	1	-
ConversionIs	0	-
DaysInDayType	7	-
DaySplit	0.00137	-
DepreciationMethod	1	-
DiscountRate	5	Ratio
EmissionActivityRatio	0	kton/PJ
EmissionPenalty	0	\$/ton of emission
FixedCost	999	M\$/GW
InputActivityRatio	0	Ratio
MinStorageCharge	0	Ratio
ModelPeriodEmissionLimit	999999	ton
ModelPeriodExogenousEmission	0	ton
OperationalLife	1	Number of years
OperationalLifeStorage	0	
OutputActivityRatio	0	Ratio
REMinProductionTarget	0	Ratio

ReserveMargin	0	
ReserveMarginTagFuel	0	
ReserveMarginTagTechnology	0	
ResidualCapacity	0	GW
ResidualStorageCapacity	0	
RETagFuel	0	
RETagTechnology	0	
SpecifiedAnnualDemand	0	PJ
SpecifiedDemandProfile	0	Ratio
StorageLevelStart	0	
StorageMaxChargeRate	0	
StorageMaxDischargeRate	0	
TechnologyFromStorage	0	
TechnologyToStorage	0	
TotalAnnualMaxCapacity	99999	GW
TotalAnnualMaxCapacityInvestment	99999	GW
TotalTechnologyAnnualActivityLowerLimit	0	PJ
TotalTechnologyAnnualActivityUpperLimit	999999	PJ
TotalTechnologyModelPeriodActivityLowerLimit	0	
TotalTechnologyModelPeriodActivityUpperLimit	999999	

rLimit		
TradeRoute	0	Binary
VariableCost	0.00001	M\$/PJ
YearSplit	0.25	Ratio

Table 7: Residual capacity input 2010-2019 - additional data for scenario C and D

Technology	Year									
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
ETHYDMS02	0,751	0,751	0,751	0,751	0,751	0,751	0,751	0,751	0,751	0,751
ETHYDMS03	0,143	0,143	0,143	0,143	0,143	0,143	0,143	0,143	0,143	0,143
ETHYDMS01	0	0	0	0	0	0	0	0	0	0
ETGOCVP00	0,005	0,005	0,005	0,005	0,005	0,005	0,005	0,005	0,005	0,005

Table 8: Residual capacity input 2020-2029 - additional data for scenario C and D

Technology	Year									
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
ETHYDMS02	0,751	0,751	0,751	0,751	0,751	0,751	0,751	0,751	0,751	0,751
ETHYDMS03	0,143	0,143	0,143	0,143	0,143	0,143	0,143	0,143	0,143	0,143
ETHYDMS01	0	0	0	0	0	0	0	0	0	0
ETGOCVP00	0,005	0,005	0,005	0,005	0,005	0,005	0,005	0,005	0,005	0,005

Table 9: Residual capacity input 2030-2040 - additional data for scenario C and D

Technology	Year										
	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
ETHYDMS02	0,751	0,751	0,751	0,751	0,751	0,751	0,711	0,711	0,711	0,711	0,711
ETHYDMS03	0,143	0,143	0,143	0,143	0,143	0,143	0,143	0,143	0,143	0,143	0,132
ETHYDMS01	0	0	0	0	0	0	0	0	0	0	0
ETGOCVP00	0,005	0,005	0,005	0	0	0	0	0	0	0	0

Table 10: TotalAnnualMinCapacityInvetsment input 2010-2019 - additional data for scenario C and D

Technology	Year									
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
ETHYDMS02	0,42	0	0	0,09	0,465	1,416	1,094	3,337	1,913	0
ETHYDMS03	0,444	0	0	0	0	0	0	0	0	0
ETHYDMS01	0	0	0	0	0,005	0	0	0	0	0
ETWI25P00	0	0	0,0405	0	0,045	0,0765	0	0,3	0,15	0
ETWI30P00	0	0	0,0405	0	0,045	0,0765	0	0,3	0,15	0
ETGOCVP00	0	0	0	0	0	0,002	0	0	0,195	0,1
ETBMCHP00	0	0	0	0,026	0	0	0,396	0,319	0	0,04
ETDSRCF12	0	0	0	0,000333333	0,026	0	0	0	0	0
ETDSRCFR1	0	0	0	0,000333333	0,026	0	0	0	0	0
ETDSRCFU1	0	0	0	0,000333333	0,026	0	0	0	0	0

ETSOC1P00	0	0	0	0	0	0	0	0,075	0	0
ETSOC2P00	0	0	0	0	0	0	0	0,075	0	0
ETSOC3P00	0	0	0	0	0	0	0	0,075	0	0
ETSOU1P03	0	0	0	0	0	0	0	0,075	0	0

Table 11: TotalAnnualMinCapacityInvetsment input 2020-2029 - additional data for scenario C and D

Technology	Year									
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
ETHYDMS02	3,739	0	0	0	0	0	0	0	0	0
ETHYDMS03	0	0	0	0	0	0	0	0	0	0
ETHYDMS01	0	0	0	0	0	0	0	0	0	0
ETWI25P00	0	0	0	0	0	0	0	0	0	0
ETWI30P00	0	0	0	0	0	0	0	0	0	0
ETGOCVP00	0,2	0	0	0	0	0	0	0	0	0
ETBMCHP00	0	0	0	0	0	0	0	0	0	0
ETDSRCF12	0	0	0	0	0	0	0	0	0	0
ETDSRCFR1	0	0	0	0	0	0	0	0	0	0
ETDSRCFU1	0	0	0	0	0	0	0	0	0	0
ETSOC1P00	0	0	0	0	0	0	0	0	0	0
ETSOC2P00	0	0	0	0	0	0	0	0	0	0
ETSOC3P00	0	0	0	0	0	0	0	0	0	0

ETSOU1P03	0	0	0	0	0	0	0	0	0	0
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Table 12: TotalAnnualMinCapacityInvetsment input 2030-2040 - additional data for scenario C and D

Technology	Year										
	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
ETHYDMS02	0	0	0	0	0	0	0	0	0	0	0
ETHYDMS03	0	0	0	0	0	0	0	0	0	0	0
ETHYDMS01	0	0	0	0	0	0	0	0	0	0	0
ETWI25P00	0	0	0	0	0	0	0	0	0	0	0
ETWI30P00	0	0	0	0	0	0	0	0	0	0	0
ETGOCVP00	0	0	0	0	0	0	0	0	0	0	0
ETBMCHP00	0	0	0	0	0	0	0	0	0	0	0
ETDSRCF12	0	0	0	0	0	0	0	0	0	0	0
ETDSRCFR1	0	0	0	0	0	0	0	0	0	0	0
ETDSRCFU1	0	0	0	0	0	0	0	0	0	0	0
ETSOC1P00	0	0	0	0	0	0	0	0	0	0	0
ETSOC2P00	0	0	0	0	0	0	0	0	0	0	0
ETSOC3P00	0	0	0	0	0	0	0	0	0	0	0
ETSOU1P03	0	0	0	0	0	0	0	0	0	0	0

Table 13: SpecifiedAnnualDemand input 2010-2019, scenario D

Distribution line	Year									
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
ETEL3I	14,4	27,18	39,96	52,74	65,52	78,3	91,08	103,6530555	115,7875285	127,4629034
ETEL3R	0	0	0	0	0	0	0	0	0	0
ETEL3U	0	0	0	0	0	0	0	0,206944503	0,85247146	1,957096636

Table 14: SpecifiedAnnualDemand input 2020-2029, scenario D

Distribution line	Year									
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
ETEL3I	138,6727318	133,6791614	138,8915536	143,4495071	147,4654625	151,0272258	154,2058237	157,0588978	159,6328202	161,9657655
ETEL3R	0	16,54000332	22,41630442	28,80050243	35,60589169	42,76375967	50,21915213	57,92742558	65,8524033	73,96449248
ETEL3U	3,527268237	4,760835314	6,452141935	8,28999048	10,2486458	12,30901454	14,45502413	16,67367666	18,95477649	21,28974198

Table 15: SpecifiedAnnualDemand input 2030-2040, scenario D

Distribution line	Year										
	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
ETEL3I	164,08 95921	168,94 27739	173,7 70704 8	178,5 57203 5	183,29 50624	187,97 63308	192,5 9787 65	197,1 5744 74	201,65 54761	206,0 89774 4	210,4 60108 5
ETEL3R	82,238 91543	87,420 48932	92,50 11966 1	97,49 21363 9	102,40 62512	107,24 69808	112,0 2101 74	116,7 3426 88	121,39 1063	125,9 67209 8	130,4 68533 4
ETEL3U	23,671 49245	26,416 73674	29,28 80985 5	32,29 06601 3	35,418 6864	38,676 68845	42,06 1106 1	45,56 8283 73	49,193 46092	52,96 30157 4	56,87 13581

Table 16: Additional technologies in MoManI, scenario E

Naming	Explanation
ETDSRMGFR	Diesel Genset - Mini Grid
ETHYDMGFR	Mini and Small Hydro - Mini Grid
ETSOVMGFR	Solar PV - Mini Grid
ETWIMGCFR	Wind Turbines - Mini Grid
ETDSRSAFR	Diesel Genset - Stand-Alone
ETSOVSAFR	Solar PV - Stand-Alone

Table 17: Input data for the harmonisation, scenario E

Technology	CapitalCost (M\$/GW)	Operational life (years)	Availability factor	Fixed costs (M\$/GW)	Variable costs (M\$/GW)	Input activity ratio
ETDSRMGFR	721	15	0,33	0	15.3889	3,0303
ETHYDMGFR	5 000	30	0,98	40,9649	2.0339	-
ETSOVMGFR	4 300	20	1	33	0	-
ETWIMGCFR	2 500	20	1	33	0	-
ETDSRSAFR	938	10	0,98	0	15.3889	3,5714
ETSOVSAFR	5 500	15	1	33	0	-

Table 18: Capacity factor for the additional technologies, scenario E

Technology	Capacity factor			
	SD	SN	WD	WN
ETDSRMGFR	0,9	0,9	0,9	0,9
ETHYDMGFR	0,5	0,5	0,5	0,5
ETSOVMGFR	0,2704	0	0,2704	0
ETWIMGCFR	0,169	0,169	0,169	0,169
ETDSRSAFR	0,9	0,9	0,9	0,9
ETSOVSAFR	0,2704	0	0,2704	0

Table 19: SpecifiedAnnualDemand input 2010-2019, scenario E

Distribution line	Year									
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
ETEL3E	13,813 2	15,732	18,99 36	21,91 32	24,76 19	27,61 06	25,83 90775	24,134 155	22,56233 25	21,12351
ETEL3UE	0	0	0	0	0	0	4,620 223	9,2404 45002	13,86066 75	18,48089

Table 20: SpecifiedAnnualDemand input 2020-2029, scenario E

Distribution line	Year									
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
ETEL3E	19,81 7687 49	26,43 0764 99	29,94 0442 49	34,04 5619 99	38,74 5997 49	44,04 19749 9	49,933 35249	56,4201 2999	63,50230 749	71,179984 98
ETEL3UE	23,10 1112 51	27,72 1335 01	32,34 1557 51	36,96 1780 01	41,58 2002 51	46,20 22250 1	50,822 44751	55,4426 7001	60,06289 251	64,683115 02

Table 21: SpecifiedAnnualDemand input 2030-2040, scenario E

Distribution line	Year										
	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
ETEL3E	79,45 30624 8	90,4 1203 998	101, 6707 175	113, 2289 95	125,08 30725	137,2 3275	149,67 54275	162,40 9205	175,42 93825	188,73 636	202,32 88375
ETEL3UE	69,30 33375 2	73,9 2356 002	78,5 4378 252	83,1 6400 502	87,784 22752	92,40 4450 02	97,024 67252	101,64 4895	106,26 51175	110,88 534	115,50 55625

Appendix IV - TEMBA data

Table 22: Parameters non-dependent to time slices

	CapacityToActivityUnit	OperationalLife
ETBACKSTOP	31,536	1
ETBM00I00	1	35
ETBM00X00	1	35
ETBMCHP00	31,536	30
ETCO00I00	1	35
ETCO00X00	1	35
ETCOSCP00	31,536	40
ETDS00I00	1	30
ETDS00X00	1	30
ETDSRCFI2	31,536	20
ETDSRCFR1	31,536	10
ETDSRCFU1	31,536	10
ETDSRCP03	31,536	25
ETEL00T00	31,536	60
ETEL00TDI	31,536	60
ETEL00TDR	31,536	60
ETEL00TDU	31,536	60
ETGOCVP00	31,536	25
ETHF00I00	1	25
ETHF00X00	1	25
ETHFGCP00	31,536	25
ETHYDMS01	31,536	50
ETHYDMS02	31,536	50
ETHYDMS03	31,536	50
ETNG00I00	1	100
ETNG00X00	1	30
ETNGCCP00	31,536	30
ETNGGCP00	31,536	25
ETSOC1P00	31,536	25
ETSOC2P00	31,536	25
ETSOC3P00	31,536	25
ETSOPV	1	100
ETSOU1P03	31,536	25
ETSOV1F01	31,536	25
ETSOV2F01	31,536	25
ETSOV3F01	31,536	25
ETSOV4F01	31,536	25

ETSOV5F01	31,536	25
ETSOV6F01	31,536	25
ETWI25P00	31,536	25
ETWI30P00	31,536	25

Table 23: Availability factor, 2010-2019

Availability Factor	Year									
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
ETBACKSTOP	1	1	1	1	1	1	1	1	1	1
ETBM00I00	1	1	1	1	1	1	1	1	1	1
ETBM00X00	1	1	1	1	1	1	1	1	1	1
ETBMCHP00	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93
ETCO00I00	1	1	1	1	1	1	1	1	1	1
ETCO00X00	1	1	1	1	1	1	1	1	1	1
ETCOSCP00	0,94	0,94	0,94	0,94	0,94	0,94	0,94	0,94	0,94	0,94
ETDS00I00	1	1	1	1	1	1	1	1	1	1
ETDS00X00	1	1	1	1	1	1	1	1	1	1
ETDSRCFI2	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETDSRCFR1	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETDSRCFU1	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETDSRCP03	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETEL00T00	1	1	1	1	1	1	1	1	1	1
ETEL00TDI	1	1	1	1	1	1	1	1	1	1
ETEL00TDR	1	1	1	1	1	1	1	1	1	1
ETEL00TDU	1	1	1	1	1	1	1	1	1	1
ETGOCVP00	1	1	1	1	1	1	1	1	1	1
ETHF00I00	1	1	1	1	1	1	1	1	1	1
ETHF00X00	1	1	1	1	1	1	1	1	1	1
ETHFGCP00	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETHYDMS01	1	1	1	1	1	1	1	1	1	1
ETHYDMS02	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95
ETHYDMS03	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95
ETNG00I00	1	1	1	1	1	1	1	1	1	1
ETNG00X00	1	1	1	1	1	1	1	1	1	1
ETNGCCP00	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93
ETNGGCP00	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93
ETSOC1P00	1	1	1	1	1	1	1	1	1	1
ETSOC2P00	1	1	1	1	1	1	1	1	1	1
ETSOC3P00	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93
ETSOPV	1	1	1	1	1	1	1	1	1	1

ETSOU1P03	1	1	1	1	1	1	1	1	1	1
ETSOV1F01	1	1	1	1	1	1	1	1	1	1
ETSOV2F01	1	1	1	1	1	1	1	1	1	1
ETSOV3F01	1	1	1	1	1	1	1	1	1	1
ETSOV4F01	1	1	1	1	1	1	1	1	1	1
ETSOV5F01	1	1	1	1	1	1	1	1	1	1
ETSOV6F01	1	1	1	1	1	1	1	1	1	1
ETWI25P00	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETWI30P00	0,85	0,85	0,85	0,85	0,85	0,85	0,85	0,85	0,85	0,85

Table 24: Availability factor 2020-2029

Availability Factor	Year									
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
ETBACKSTOP	1	1	1	1	1	1	1	1	1	1
ETBM00I00	1	1	1	1	1	1	1	1	1	1
ETBM00X00	1	1	1	1	1	1	1	1	1	1
ETBMCHP00	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93
ETCO00I00	1	1	1	1	1	1	1	1	1	1
ETCO00X00	1	1	1	1	1	1	1	1	1	1
ETCOSCP00	0,94	0,94	0,94	0,94	0,94	0,94	0,94	0,94	0,94	0,94
ETDS00I00	1	1	1	1	1	1	1	1	1	1
ETDS00X00	1	1	1	1	1	1	1	1	1	1
ETDSRCFI2	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETDSRCFR1	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETDSRCFU1	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETDSRCP03	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETEL00T00	1	1	1	1	1	1	1	1	1	1
ETEL00TDI	1	1	1	1	1	1	1	1	1	1
ETEL00TDR	1	1	1	1	1	1	1	1	1	1
ETEL00TDU	1	1	1	1	1	1	1	1	1	1
ETGOCVP00	1	1	1	1	1	1	1	1	1	1
ETHF00I00	1	1	1	1	1	1	1	1	1	1
ETHF00X00	1	1	1	1	1	1	1	1	1	1
ETHFGCP00	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETHYDMS01	1	1	1	1	1	1	1	1	1	1
ETHYDMS02	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95
ETHYDMS03	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95
ETNG00I00	1	1	1	1	1	1	1	1	1	1
ETNG00X00	1	1	1	1	1	1	1	1	1	1
ETNGCCP00	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93

ETNGGCP00	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93
ETSOC1P00	1	1	1	1	1	1	1	1	1	1
ETSOC2P00	1	1	1	1	1	1	1	1	1	1
ETSOC3P00	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93
ETSOPV	1	1	1	1	1	1	1	1	1	1
ETSOU1P03	1	1	1	1	1	1	1	1	1	1
ETSOV1F01	1	1	1	1	1	1	1	1	1	1
ETSOV2F01	1	1	1	1	1	1	1	1	1	1
ETSOV3F01	1	1	1	1	1	1	1	1	1	1
ETSOV4F01	1	1	1	1	1	1	1	1	1	1
ETSOV5F01	1	1	1	1	1	1	1	1	1	1
ETSOV6F01	1	1	1	1	1	1	1	1	1	1
ETWI25P00	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETWI30P00	0,85	0,85	0,85	0,85	0,85	0,85	0,85	0,85	0,85	0,85

Table 25: Availability factor 2030-2040

Availability Factor	Year										
	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
ETBACKSTOP	1	1	1	1	1	1	1	1	1	1	1
ETBM00I00	1	1	1	1	1	1	1	1	1	1	1
ETBM00X00	1	1	1	1	1	1	1	1	1	1	1
ETBMCHP00	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93
ETCO00I00	1	1	1	1	1	1	1	1	1	1	1
ETCO00X00	1	1	1	1	1	1	1	1	1	1	1
ETCOSCP00	0,94	0,94	0,94	0,94	0,94	0,94	0,94	0,94	0,94	0,94	0,94
ETDS00I00	1	1	1	1	1	1	1	1	1	1	1
ETDS00X00	1	1	1	1	1	1	1	1	1	1	1
ETDSRCFI2	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETDSRCFR1	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETDSRCFU1	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETDSRCP03	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETEL00T00	1	1	1	1	1	1	1	1	1	1	1
ETEL00TDI	1	1	1	1	1	1	1	1	1	1	1
ETEL00TDR	1	1	1	1	1	1	1	1	1	1	1
ETEL00TDU	1	1	1	1	1	1	1	1	1	1	1
ETGOCVP00	1	1	1	1	1	1	1	1	1	1	1
ETHF00I00	1	1	1	1	1	1	1	1	1	1	1
ETHF00X00	1	1	1	1	1	1	1	1	1	1	1
ETHFGCP00	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETHYDMS01	1	1	1	1	1	1	1	1	1	1	1

ETHYDMS02	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95
ETHYDMS03	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95	0,95
ETNG00I00	1	1	1	1	1	1	1	1	1	1	1
ETNG00X00	1	1	1	1	1	1	1	1	1	1	1
ETNGCCP00	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93
ETNGGCP00	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93
ETSOC1P00	1	1	1	1	1	1	1	1	1	1	1
ETSOC2P00	1	1	1	1	1	1	1	1	1	1	1
ETSOC3P00	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93	0,93
ETSOPV	1	1	1	1	1	1	1	1	1	1	1
ETSOU1P03	1	1	1	1	1	1	1	1	1	1	1
ETSOV1F01	1	1	1	1	1	1	1	1	1	1	1
ETSOV2F01	1	1	1	1	1	1	1	1	1	1	1
ETSOV3F01	1	1	1	1	1	1	1	1	1	1	1
ETSOV4F01	1	1	1	1	1	1	1	1	1	1	1
ETSOV5F01	1	1	1	1	1	1	1	1	1	1	1
ETSOV6F01	1	1	1	1	1	1	1	1	1	1	1
ETWI25P00	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9	0,9
ETWI30P00	0,85	0,85	0,85	0,85	0,85	0,85	0,85	0,85	0,85	0,85	0,85

Table 26: Availability factor that are constant throughout a year 2010-2040

ETBACKSTOP	1
ETBM00I00	1
ETBM00X00	1
ETBMCHP00	0,93
ETCO00I00	1
ETCO00X00	1
ETCOSCP00	0,94
ETDS00I00	1
ETDS00X00	1
ETDSRCFI2	0,9
ETDSRCFR1	0,9
ETDSRCFU1	0,9
ETDSRCP03	0,9
ETEL00T00	1
ETEL00TDI	1
ETEL00TDR	1
ETEL00TDU	1
ETGOCVP00	1
ETHF00I00	1

ETHF00X00	1
ETHFGCP00	0,9
ETHYDMS01	1
ETHYDMS02	0,95
ETHYDMS03	0,95
ETNG00I00	1
ETNG00X00	1
ETNGCCP00	0,93
ETNGGCP00	0,93
ETSOC1P00	1
ETSOC2P00	1
ETSOC3P00	0,93
ETSOPV	1
ETSOU1P03	1
ETSOV1F01	1
ETSOV2F01	1
ETSOV3F01	1
ETSOV4F01	1
ETSOV5F01	1
ETSOV6F01	1
ETWI25P00	0,9
ETWI30P00	0,85

Table 27: Capacity factors that are constant throughout a year, 2010-2040

ETBMCHP00	0,53
ETCOSCP00	0,85
ETDSRCFI2	0,9
ETDSRCFR1	0,9
ETDSRCFU1	0,9
ETDSRCP03	0,9
ETGOCVP00	0,93
ETHFGCP00	0,9
ETHYDMS01	0,25
ETHYDMS02	0,5386
ETHYDMS03	0,5386
ETNGCCP00	0,935
ETNGGCP00	0,935
ETSOC3P00	0,92
ETSOU1P03	0,5
ETWI25P00	0,25

ETWI30P00	0,3
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Table 28: Capacity factors that vary during a year, 2010-2040

Technology	Summer day	Summer night	Winter day	Winter night
ETSOC1P00	0,7	0	0,7	0
ETSOC2P00	0,7056	0,5544	0,7056	0,5544
ETSOU1P03	0,5	0	0,5	0
ETSOV1F01	0,4	0	0,4	0
ETSOV2F01	0,378	0,072	0,378	0,072
ETSOV3F01	0,2856	0,2244	0,2856	0,2244
ETSOV4F01	0,4	0	0,4	0
ETSOV5F01	0,378	0,072	0,378	0,072
ETSOV6F01	0,2856	0,2244	0,2856	0,2244

Table 29: Capital cost that are constant throughout a year, 2010-2040

ETBACKSTOP	9999
ETBM00I00	0
ETBM00X00	0
ETCO00I00	0
ETCO00X00	0
ETCOSCP00	2528,253
ETDS00I00	0
ETDS00X00	0
ETDSRCFI2	691,59825
ETDSRCFR1	726,46875
ETDSRCFU1	691,875
ETDSRCP03	780,57
ETEL00T00	365
ETEL00TDI	840,4
ETEL00TDR	4233,6
ETEL00TDU	2433,3
ETHF00I00	0
ETHF00X00	0
ETHFGCP00	1488,375
ETNG00I00	0
ETNG00X00	0
ETNGCCP00	1181,935125
ETNGGCP00	780,57
ETSOC3P00	1590,57675

ETSOPV	0
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Table 30: Capital cost that vary during a year, 2010-2019

Technology	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
ETBMCHP00	5147,96	5147,96	5147,96	5130,30	5112,64	5094,98	5077,32	5059,66	5042,00	5024,34
ETGOCVP00	5953,55	5953,55	5953,55	5933,90	5914,25	5894,59	5874,94	5855,29	5835,64	5815,98
ETHYDMS01	6231,90	6231,90	6231,90	6223,98	6216,07	6208,15	6200,24	6192,32	6184,40	6176,49
ETHYDMS02	2159,35	2159,35	2159,35	2165,12	2170,89	2176,67	2182,44	2188,22	2193,99	2199,76
ETHYDMS03	2159,35	2159,35	2159,35	2165,12	2170,89	2176,67	2182,44	2188,22	2193,99	2199,76
ETSOC1P00	5773,65	5773,65	5773,65	5578,78	5383,91	5189,04	4994,17	4799,30	4604,42	4409,55
ETSOC2P00	7845,22	7845,22	7845,22	7448,37	7051,51	6654,65	6257,80	5860,94	5464,08	5067,22
ETSOU1P03	1680,60	1680,60	1680,60	1626,32	1572,03	1517,75	1463,46	1409,18	1354,89	1300,61
ETSOV1F01	2075,85	2075,85	2075,85	2008,17	1940,48	1872,80	1805,11	1737,43	1669,74	1602,06
ETSOV2F01	4258	4258	4100,6	3943,2	3785,8	3628,4	3471	3373	3275	3177
ETSOV3F01	6275	6275	6053,2	5831,4	5609,6	5387,8	5166	5011,6	4857,2	4702,8
ETSOV4F01	2075,85	2075,85	2075,85	2008,17	1940,48	1872,80	1805,11	1737,43	1669,74	1602,06
ETSOV5F01	4258	4258	4100,6	3943,2	3785,8	3628,4	3471	3373	3275	3177
ETSOV6F01	6275	6275	6053,2	5831,4	5609,6	5387,8	5166	5011,6	4857,2	4702,8
ETWI25P00	2191,67	2191,67	2191,67	2169,84	2148,00	2126,17	2104,34	2082,50	2060,67	2038,84
ETWI30P00	2191,67	2191,67	2191,67	2169,84	2148,00	2126,17	2104,34	2082,50	2060,67	2038,84

Table 31: Capital cost that vary during a year, 2020-2029

Table 32: Capital cost that vary during a year, 2030-2040

Technology	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
ETBMCHP00	5006,7	4989,7	4972,7	4955,7	4938,8	4921,8	4904,8	4887,8	4870,8	4853,8
ETGOCVP00	5796,3	5762,9	5729,5	5696,1	5662,6	5629,2	5595,8	5562,4	5528,9	5495,5
ETHYDMS01	6168,6	6164,5	6160,4	6156,3	6152,3	6148,2	6144,1	6140,0	6136,0	6131,9
ETHYDMS02	2205,5	2214,8	2224,0	2233,2	2242,5	2251,7	2261,0	2270,2	2279,4	2288,7
ETHYDMS03	2205,5	2214,8	2224,0	2233,2	2242,5	2251,7	2261,0	2270,2	2279,4	2288,7
ETSOC1P00	4214,7	4149,1	4083,4	4017,8	3952,2	3886,6	3821,0	3755,4	3689,8	3624,1
ETSOC2P00	4670,4	4591,2	4512,0	4432,9	4353,7	4274,6	4195,4	4116,2	4037,1	3957,9
ETSOU1P03	1246,3	1230,3	1214,2	1198,2	1182,1	1166,1	1150,0	1134,0	1117,9	1101,9
ETSOV1F01	1534,4	1514,3	1494,2	1474,2	1454,1	1434,0	1413,9	1393,9	1373,8	1353,7
ETSOV2F01	3079,0	2981,0	2926,5	2872,0	2817,5	2763,0	2708,5	2654,0	2599,5	2545,0
ETSOV3F01	4548,4	4394,0	4313,6	4233,2	4152,8	4072,4	3992,0	3911,6	3831,2	3750,8
ETSOV4F01	1534,4	1514,3	1494,2	1474,2	1454,1	1434,0	1413,9	1393,9	1373,8	1353,7
ETSOV5F01	3079,0	2981,0	2926,5	2872,0	2817,5	2763,0	2708,5	2654,0	2599,5	2545,0
ETSOV6F01	4548,4	4394,0	4313,6	4233,2	4152,8	4072,4	3992,0	3911,6	3831,2	3750,8
ETWI25P00	2017,0	2007,6	1998,1	1988,7	1979,3	1969,8	1960,4	1951,0	1941,5	1932,1
ETWI30P00	2017,0	2007,6	1998,1	1988,7	1979,3	1969,8	1960,4	1951,0	1941,5	1932,1

Technology	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
ETCOSCP00	2528,3	2528,3	2528,3	2528,3	2528,3	2528,3	2528,3	2528,3	2528,3	2528,3	2528,3
ETGOCVP00	5462,1	5428,7	5395,2	5361,8	5328,4	5295,0	5295,0	5295,0	5295,0	5295,0	5295,0
ETHYDMS01	6127,8	6123,7	6119,7	6115,6	6111,5	6107,4	6107,4	6107,4	6107,4	6107,4	6107,4
ETHYDMS02	2297,9	2307,2	2316,4	2325,6	2334,9	2344,1	2344,1	2344,1	2344,1	2344,1	2344,1
ETHYDMS03	2297,9	2307,2	2316,4	2325,6	2334,9	2344,1	2344,1	2344,1	2344,1	2344,1	2344,1
ETSOC1P00	3558,5	3492,9	3427,3	3361,7	3296,1	3230,5	3230,5	3230,5	3230,5	3230,5	3230,5
ETSOC2P00	3878,8	3799,6	3720,4	3641,3	3562,1	3482,9	3482,9	3482,9	3482,9	3482,9	3482,9
ETSOU1P03	1085,8	1069,8	1053,7	1037,7	1021,6	1005,6	1005,6	1005,6	1005,6	1005,6	1005,6
ETSOV1F01	1333,6	1313,6	1293,5	1273,4	1253,3	1233,3	1233,3	1233,3	1233,3	1233,3	1233,3
ETSOV2F01	2490,5	2436,0	2413,8	2391,6	2369,4	2347,2	2325,0	2302,8	2280,6	2258,4	2236,2
ETSOV3F01	3670,4	3590,0	3557,3	3524,6	3491,9	3459,2	3426,5	3393,8	3361,1	3328,4	3295,7
ETSOV4F01	1333,6	1313,6	1293,5	1273,4	1253,3	1233,3	1233,3	1233,3	1233,3	1233,3	1233,3
ETSOV5F01	2490,5	2436,0	2413,8	2391,6	2369,4	2347,2	2325,0	2302,8	2280,6	2258,4	2236,2
ETSOV6F01	3670,4	3590,0	3557,3	3524,6	3491,9	3459,2	3426,5	3393,8	3361,1	3328,4	3295,7
ETWI25P00	1922,7	1913,2	1903,8	1894,3	1884,9	1875,5	1875,5	1875,5	1875,5	1875,5	1875,5
ETWI30P00	1922,7	1913,2	1903,8	1894,3	1884,9	1875,5	1875,5	1875,5	1875,5	1875,5	1875,5

Table 33: Emission Activity Ratio, 2010-2040

ETCO00I00	0,0893
ETCO00X00	0,0893
ETDS00I00	0,0693
ETDS00X00	0,0693
ETHF00I00	0,0747
ETHF00X00	0,0747
ETNG00I00	0,0503
ETNG00X00	0,0503

Table 34: Fixed cost, 2010-2040

ETBACKSTOP	9999
ETBM00I00	0
ETBM00X00	0
ETCO00I00	0
ETCO00X00	0
ETCOSCP00	26,74815973
ETDS00I00	0
ETDS00X00	0
ETDSRCFI2	0
ETDSRCFR1	0

ETDSRCFU1	0
ETDSRCP03	8
ETEL00T00	0
ETEL00TDI	0
ETEL00TDR	0
ETEL00TDU	0
ETHF00I00	0
ETHF00X00	0
ETHFGCP00	0
ETHYDMS01	40,96489678
ETHYDMS02	25,91131873
ETNG00I00	0
ETNG00X00	0
ETNGCCP00	1,323058618
ETNGGCP00	8,328946435
ETSOC3P00	0
ETSOPV	0
ETSOV2F01	0
ETSOV3F01	0
ETSOV5F01	0
ETSOV6F01	0

Table 35: Fixed cost, 2010-2019

Technology	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
ETBMCHP00	58,64	58,64	58,64	58,32	57,99	57,67	57,34	57,02	56,69	56,37
ETGOCVP00	49	49	49	48,875	48,75	48,63	48,50	48,38	48,25	48,13
ETHYDMS03	22,15	22,15	22,15	22,15	22,15	22,15	22,15	22,15	22,15	22,15
ETSOC1P00	200	200	200	193	186	180	173	167	160	154
ETSOC2P00	61	61	61	59	57	54	52	49	47	44
ETSOU1P03	25	25	25	24,75	24,5	24,25	24	23,75	23,5	23,25
ETSOV1F01	33	33	33	32,75	32,5	32,25	32	31,75	31,5	31,25
ETSOV4F01	33	33	33	32,75	32,5	32,25	32	31,75	31,5	31,25
ETWI25P00	37	37	37	36,63	36,25	35,88	35,50	35,13	34,75	34,38
ETWI30P00	37	37	37	36,63	36,25	35,88	35,50	35,13	34,75	34,38

Table 36: Fixed cost, 2020-2029

Technology	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
ETBMCHP00	56,04	55,93	55,82	55,72	55,61	55,50	55,39	55,28	55,18	55,07
ETGOCVP00	48	47,73	47,47	47,20	46,93	46,67	46,40	46,13	45,87	45,60
ETHYDMS03	22,15	22,30	22,45	22,59	22,74	22,89	23,04	23,18	23,33	23,48
ETSOC1P00	147,3	145,0	142,8	140,6	138,4	136,2	134,0	131,7	129,5	127,3
ETSOC2P00	41,8	41,8	41,8	41,8	41,8	41,8	41,8	41,8	41,8	41,8
ETSOU1P03	23,0	22,9	22,9	22,8	22,7	22,7	22,6	22,5	22,5	22,4
ETSOV1F01	31,0	30,9	30,7	30,6	30,5	30,3	30,2	30,1	29,9	29,8
ETSOV4F01	31,0	30,9	30,7	30,6	30,5	30,3	30,2	30,1	29,9	29,8
ETWI25P00	34,0	33,9	33,9	33,8	33,7	33,7	33,6	33,5	33,5	33,4
ETWI30P00	34,0	33,9	33,9	33,8	33,7	33,7	33,6	33,5	33,5	33,4

Table 37: Fixed cost, 2030-2040

Technology	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
ETBMCHP00	54,96	54,85	54,74	54,63	54,53	54,42	54,42	54,42	54,42	54,42	54,42
ETGOCVP00	45,33	45,07	44,80	44,53	44,27	44,00	44,00	44,00	44,00	44,00	44,00
ETHYDMS03	23,62	23,77	23,92	24,06	24,21	24,36	24,36	24,36	24,36	24,36	24,36
ETSOC1P00	125,08	122,87	120,65	118,43	116,22	114,00	114,00	114,00	114,00	114,00	114,00
ETSOC2P00	41,78	41,78	41,78	41,78	41,78	41,78	41,78	41,78	41,78	41,78	41,78
ETSOU1P03	22,33	22,27	22,20	22,13	22,07	22,00	22,00	22,00	22,00	22,00	22,00
ETSOV1F01	29,67	29,53	29,40	29,27	29,13	29,00	29,00	29,00	29,00	29,00	29,00
ETSOV4F01	29,67	29,53	29,40	29,27	29,13	29,00	29,00	29,00	29,00	29,00	29,00
ETWI25P00	33,33	33,27	33,20	33,13	33,07	33,00	33,00	33,00	33,00	33,00	33,00
ETWI30P00	33,33	33,27	33,20	33,13	33,07	33,00	33,00	33,00	33,00	33,00	33,00

Table 38: Input Activity Ratio, 2010-2040

Technology	Fuel	
ETBMCHP00	ETBIOMASS	2,631578
ETCOSCP00	ETCOAL	2,7027
ETDSRCFI2	ETDIESEL	2,85714
ETDSRCFR1	ETDIESEL	4,76
ETDSRCFU1	ETDIESEL	4,76
ETDSRCP03	ETDIESEL	2,85714
ETEL00T00	ETEL1	1
ETEL00TDI	ETEL2	1
ETEL00TDR	ETEL2	1
ETEL00TDU	ETEL2	1
ETHFGCP00	ETOIL	2,85714
ETNGCCP00	ETGAS	2,083333

ETNGGCP00	ETGAS	3,33
ETSOC3P00	ETGAS	1,886792
ETSOU1P03	ETSOLARPV	1
ETSOV1F01	ETSOLARPV	1
ETSOV2F01	ETSOLARPV	1
ETSOV3F01	ETSOLARPV	1
ETSOV4F01	ETSOLARPV	1
ETSOV5F01	ETSOLARPV	1
ETSOV6F01	ETSOLARPV	1
ETBACKSTOP	ETEL1	1

Table 39: Output Activity Ratio, 2010-2040

Technology	Fuel	
ETBACKSTOP	ETEL1	1
ETBM00I00	ETBIOMASS	1
ETBM00X00	ETBIOMASS	1
ETBMCHP00	ETEL1	1
ETCO00I00	ETCOAL	1
ETCO00X00	ETCOAL	1
ETCOSCP00	ETEL1	1
ETDS00I00	ETDIESEL	1
ETDS00X00	ETDIESEL	1
ETDSRCFI2	ETEL3I	1
ETDSRCFR1	ETEL3R	1
ETDSRCFU1	ETEL3U	1
ETDSRCP03	ETEL1	1
ETEL00T00	ETEL2	0,9663
ETEL00TDI	ETEL3I	0,96
ETEL00TDR	ETEL3R	0,88
ETEL00TDU	ETEL3U	0,956
ETGOCVP00	ETEL1	1
ETHF00I00	ETOIL	1
ETHF00X00	ETOIL	1
ETHFGCP00	ETEL1	1
ETHYDMS01	ETEL1	1
ETHYDMS02	ETEL1	1
ETHYDMS03	ETEL1	1
ETNG00I00	ETGAS	1
ETNG00X00	ETGAS	1
ETNGCCP00	ETEL1	1

ETNGGCP00	ETEL1	1
ETSOC1P00	ETEL1	1
ETSOC2P00	ETEL1	1
ETSOC3P00	ETEL1	1
ETSOPV	ETSOLARPV	1
ETSOU1P03	ETEL1	1
ETSOV1F01	ETEL3R	1
ETSOV2F01	ETEL3R	1
ETSOV3F01	ETEL3R	1
ETSOV4F01	ETEL3U	1
ETSOV5F01	ETEL3U	1
ETSOV6F01	ETEL3U	1
ETWI25P00	ETEL1	1
ETWI30P00	ETEL1	1

Table 40: Residual capacity, 2010-2019

Technology	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
ETBMCHP00	0,001	0,001	0,001	0,001	0,001	0,001	0,0009	0,0008	0,0007	0,0006
ETCOSCP00	0	0	0	0	0	0	0	0	0	0
ETDSRCFI2	0	0	0	0	0	0	0	0	0	0
ETDSRCFR1	0	0	0	0	0	0	0	0	0	0
ETDSRCFU1	0	0	0	0	0	0	0	0	0	0
ETDSRCP03	0,108034	0,108034	0,108034	0,108034	0,108034	0,108034	0,1061321	0,1043742	0,1024613	0,0992234
ETEL00T00	0	0	0	0	0	0	0	0	0	0
ETEL00TDI	0	0	0	0	0	0	0	0	0	0
ETEL00TDR	0	0	0	0	0	0	0	0	0	0
ETEL00TDU	0	0	0	0	0	0	0	0	0	0
ETGOCVP00	0,0085	0,0085	0,0085	0,0085	0,0085	0,0085	0,0085	0,0085	0,0085	0,0085
ETHFGCP00	0	0	0	0	0	0	0	0	0	0
ETHYDMS01	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008
ETHYDMS02	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512
ETHYDMS03	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904
ETNGCCP00	0	0	0	0	0	0	0	0	0	0
ETNGGCP00	0	0	0	0	0	0	0	0	0	0
ETSOC1P00	0	0	0	0	0	0	0	0	0	0
ETSOC2P00	0	0	0	0	0	0	0	0	0	0
ETSOC3P00	0	0	0	0	0	0	0	0	0	0
ETSOU1P03	0,00015	0,00015	0,00015	0,00015	0,00015	0,00015	0,00015	0,00015	0,00015	0,00015
ETSOV1F01	0	0	0	0	0	0	0	0	0	0

ETSOV2F01	0	0	0	0	0	0	0	0	0	0
ETSOV3F01	0	0	0	0	0	0	0	0	0	0
ETSOV4F01	0	0	0	0	0	0	0	0	0	0
ETSOV5F01	0	0	0	0	0	0	0	0	0	0
ETSOV6F01	0	0	0	0	0	0	0	0	0	0
ETWI25P00	0	0	0	0	0	0	0	0	0	0
ETWI30P00	0	0	0	0	0	0	0	0	0	0

Table 41: Residual capacity, 2020-2029

Technology	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
ETBMCHP00	0,0005	0,0004	0,0003	0,0002	0,0001	0	0	0	0	0
ETCOSCP00	0	0	0	0	0	0	0	0	0	0
ETDSRCFI2	0	0	0	0	0	0	0	0	0	0
ETDSRCFR1	0	0	0	0	0	0	0	0	0	0
ETDSRCFU1	0	0	0	0	0	0	0	0	0	0
ETDSRCP03	0,0953815	0,0920126	0,0904097	0,0888068	0,0848039	0,083201	0,083201	0,083201	0,002	0,002
ETEL00T00	0	0	0	0	0	0	0	0	0	0
ETEL00TDI	0	0	0	0	0	0	0	0	0	0
ETEL00TDR	0	0	0	0	0	0	0	0	0	0
ETEL00TDU	0	0	0	0	0	0	0	0	0	0
ETGOCVP00	0,0085	0,0085	0,0085	0,0046	0,0046	0,0046	0	0	0	0
ETHFGCP00	0	0	0	0	0	0	0	0	0	0
ETHYDMS01	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008
ETHYDMS02	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512
ETHYDMS03	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904
ETNGCCP00	0	0	0	0	0	0	0	0	0	0
ETNGGCP00	0	0	0	0	0	0	0	0	0	0
ETSOC1P00	0	0	0	0	0	0	0	0	0	0
ETSOC2P00	0	0	0	0	0	0	0	0	0	0
ETSOC3P00	0	0	0	0	0	0	0	0	0	0
ETSOU1P03	0,00015	0,00015	0,00015	0,00015	0,00015	0,00015	0,00015	0,00015	0,00015	0,00015
ETSOV1F01	0	0	0	0	0	0	0	0	0	0
ETSOV2F01	0	0	0	0	0	0	0	0	0	0
ETSOV3F01	0	0	0	0	0	0	0	0	0	0
ETSOV4F01	0	0	0	0	0	0	0	0	0	0
ETSOV5F01	0	0	0	0	0	0	0	0	0	0
ETSOV6F01	0	0	0	0	0	0	0	0	0	0
ETWI25P00	0	0	0	0	0	0	0	0	0	0
ETWI30P00	0	0	0	0	0	0	0	0	0	0

Table 42: Residual capacity, 2030-2040

Technology	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
ETBMCHP00	0	0	0	0	0	0	0	0	0	0	0
ETCOSCP00	0	0	0	0	0	0	0	0	0	0	0
ETDSRCFI2	0	0	0	0	0	0	0	0	0	0	0
ETDSRCFR1	0	0	0	0	0	0	0	0	0	0	0
ETDSRCFU1	0	0	0	0	0	0	0	0	0	0	0
ETDSRCP03	0,002	0,002	0,002	0,002	0	0	0	0	0	0	0
ETEL00T00	0	0	0	0	0	0	0	0	0	0	0
ETEL00TDI	0	0	0	0	0	0	0	0	0	0	0
ETEL00TDR	0	0	0	0	0	0	0	0	0	0	0
ETEL00TDU	0	0	0	0	0	0	0	0	0	0	0
ETGOCVP00	0	0	0	0	0	0	0	0	0	0	0
ETHFGCP00	0	0	0	0	0	0	0	0	0	0	0
ETHYDMS01	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008	0,003008
ETHYDMS02	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512	0,08512
ETHYDMS03	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904	1,0904
ETNGCCP00	0	0	0	0	0	0	0	0	0	0	0
ETNGGCP00	0	0	0	0	0	0	0	0	0	0	0
ETSOC1P00	0	0	0	0	0	0	0	0	0	0	0
ETSOC2P00	0	0	0	0	0	0	0	0	0	0	0
ETSOC3P00	0	0	0	0	0	0	0	0	0	0	0
ETSOU1P03	0,00015	0,00015	0,00015	0	0	0	0	0	0	0	0
ETSOV1F01	0	0	0	0	0	0	0	0	0	0	0
ETSOV2F01	0	0	0	0	0	0	0	0	0	0	0
ETSOV3F01	0	0	0	0	0	0	0	0	0	0	0
ETSOV4F01	0	0	0	0	0	0	0	0	0	0	0
ETSOV5F01	0	0	0	0	0	0	0	0	0	0	0
ETSOV6F01	0	0	0	0	0	0	0	0	0	0	0
ETWI25P00	0	0	0	0	0	0	0	0	0	0	0
ETWI30P00	0	0	0	0	0	0	0	0	0	0	0

Table 43: Specified Annual Demand, 2010-2040

Year	ETEL3I	ETEL3R	ETEL3U
2010	13,8132	0	0

2011	15,732	0	0
2012	18,9936	0	0
2013	21,9132	0	0
2014	24,7619	0	0
2015	27,6106	0	0
2016	30,4593	0	0
2017	33,3081	0	0,0665
2018	36,1568	0	0,2662
2019	39,0055	0	0,5989
2020	41,8542	0	1,0646
2021	46,7093	5,7793	1,6635
2022	51,5644	8,3222	2,3954
2023	56,4195	11,3274	3,2605
2024	61,2746	14,7949	4,2585
2025	66,1297	18,7248	5,3897
2026	70,9847	23,1171	6,654
2027	75,8398	27,9717	8,0513
2028	80,6949	33,2886	9,5817
2029	85,55	39,0679	11,2452
2030	90,4051	45,3095	13,0418
2031	98,1799	50,8038	15,3519
2032	105,9548	56,4016	17,8581
2033	113,7296	62,0963	20,5671
2034	121,5045	67,8841	23,4787
2035	129,2793	73,7583	26,5996
2036	137,0541	79,715	29,931
2037	144,829	85,7513	33,4738
2038	152,6038	91,8633	37,2274
2039	160,3786	98,0274	41,2157
2040	168,1535	104,2418	45,4391

Table 44: Specified Demand Profile, 2010-2019

Fuel	Time slice	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
ETEL3I	SD	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262
ETEL3I	SN	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240
ETEL3I	WD	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257
ETEL3I	WN	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241
ETEL3R	SD	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262
ETEL3R	SN	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240
ETEL3R	WD	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257

ETEL3R	WN	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241
ETEL3U	SD	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262
ETEL3U	SN	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240
ETEL3U	WD	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257
ETEL3U	WN	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241

Table 45: Specified Demand Profile, 2020-2029

Fuel	Time slice	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
ETEL3I	SD	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262
ETEL3I	SN	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240
ETEL3I	WD	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257
ETEL3I	WN	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241
ETEL3R	SD	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262
ETEL3R	SN	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240
ETEL3R	WD	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257
ETEL3R	WN	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241
ETEL3U	SD	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262	0,262
ETEL3U	SN	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240	0,240
ETEL3U	WD	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257	0,257
ETEL3U	WN	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241	0,241

Table 46: Specified Demand Profile, 2030-2040

Fuel	Time slice	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
ETEL3I	SD	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26
ETEL3I	SN	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24
ETEL3I	WD	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26
ETEL3I	WN	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24
ETEL3R	SD	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26
ETEL3R	SN	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24
ETEL3R	WD	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26
ETEL3R	WN	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24
ETEL3U	SD	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26
ETEL3U	SN	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24
ETEL3U	WD	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26	0,26
ETEL3U	WN	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24	0,24

Table 47: Total Annual Max Capacity

Technology	
ETGOCVP00	5

ETHFGCP00	99999
ETHYDMS01	0,003008
ETHYDMS02	0,08512
ETHYDMS03	1,5504
ETNGCCP00	99999
ETNGGCP00	99999
ETSOC1P00	99999
ETSOC2P00	99999
ETSOC3P00	99999
ETSOPV	99999
ETSOU1P03	99999
ETSOV1F01	99999
ETSOV2F01	99999
ETSOV3F01	99999
ETSOV4F01	99999
ETSOV5F01	99999
ETSOV6F01	99999
ETWI25P00	99999
ETWI30P00	99999

Table 48: Total Annual Max Capacity Investment, 2010-2019

Technology	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
ETBMCHP00	0	0	0	0	0	0	0	0	0	0
ETCOSCP00	0	0	0	0	0	0	0	0	0	0
ETDSRCFI2	99999	99999	99999	99999	99999	99999	99999	99999	99999	99999
ETDSRCFR1	99999	99999	99999	99999	99999	99999	99999	99999	99999	99999
ETDSRCFU1	99999	99999	99999	99999	99999	99999	99999	99999	99999	99999
ETDSRCP03	0	0	0	0	0	0	0	0	0	0
ETGOCVP00	0	0	0	0	0	0,06	0,01	0,09	0	0
ETHFGCP00	0	0	0	0	0	0	0	0	0	0
ETHYDMS01	0	0	0	0	0	0	0,0066	0	0	0
ETHYDMS02	0	0,097	0	0	0	0	0	0	0	0
ETHYDMS03	0,46	0	0	0	0	0,187	1,9371	0,7	4,05	0
ETNGCCP00	0	0	0	0	0	0	0	0	0	0
ETNGGCP00	0	0	0	0	0	0	0	0	0	0
ETSOC1P00	0	0	0	0	0	0	0	0	0	0
ETSOC2P00	0	0	0	0	0	0	0	0	0	0
ETSOC3P00	0	0	0	0	0	0	0	0	0	0
ETSOU1P03	0,0001	0	0	0	0	0	0	0	0	0
ETSOV1F01	0	0	0	0	0	0	0	0	0	0

ETSOV2F01	0	0	0	0	0	0	0	0	0	0,05
ETSOV3F01	0	0	0	0	0	0	0	0	0	0,05
ETSOV4F01	0	0	0	0	0	0	0	0	0	0
ETSOV5F01	0	0	0	0	0	0	0	0	0	0,05
ETSOV6F01	0	0	0	0	0	0	0	0	0	0,05
ETWI25P00	0	0	0	0	0	0	0	0	0	0
ETWI30P00	0	0,03	0,051	0,0902	0	0,051	0	0	0	0

Table 49: Total Annual Max Capacity Investment, 2020-2029

Technology	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
ETBMCHP00	0	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETCOSCP00	0	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETDSRCFI2	99999	99999	99999	99999	99999	99999	99999	99999	99999	99999
ETDSRCFR1	99999	99999	99999	99999	99999	99999	99999	99999	99999	99999
ETDSRCFU1	99999	99999	99999	99999	99999	99999	99999	99999	99999	99999
ETDSRCP03	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETGOCVP00	0	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETHEGCP00	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETHYDMS01	0	0	0	0	0	0	0	0	0	0
ETHYDMS02	0	0,065	0,065	0,065	0,065	0,065	0,065	0,065	0,065	0,065
ETHYDMS03	0	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETNGCCP00	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETNGGCP00	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETSOC1P00	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETSOC2P00	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETSOC3P00	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETSOU1P03	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETSOV1F01	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETSOV2F01	0,1707	0,2915	0,4122	0,5329	0,6537	0,7744	0,8951	1,0159	1,1366	1,2573
ETSOV3F01	0,1707	0,2915	0,4122	0,5329	0,6537	0,7744	0,8951	1,0159	1,1366	1,2573
ETSOV4F01	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETSOV5F01	0,1707	0,2915	0,4122	0,5329	0,6537	0,7744	0,8951	1,0159	1,1366	1,2573
ETSOV6F01	0,1707	0,2915	0,4122	0,5329	0,6537	0,7744	0,8951	1,0159	1,1366	1,2573
ETWI25P00	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETWI30P00	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5

Table 50: Total Annual Max Capacity Investment, 2030-2040

Technology	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
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ETBMCHP00	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3
ETCOSCP00	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3
ETDSRCFI2	99999	99999	99999	99999	99999	99999	99999	99999	99999	99999	99999
ETDSRCFR1	99999	99999	99999	99999	99999	99999	99999	99999	99999	99999	99999
ETDSRCFU1	99999	99999	99999	99999	99999	99999	99999	99999	99999	99999	99999
ETDSRCP03	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3
ETGOCVP00	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3
ETHFGCP00	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3
ETHYDMS01	0	0	0	0	0	0	0	0	0	0	0
ETHYDMS02	0,065	0,065	0,065	0,065	0,065	0,065	0,065	0,065	0,065	0,065	0,065
ETHYDMS03	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5	0,5
ETNGCCP00	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3
ETNGGCP00	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3
ETSOC1P00	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3
ETSOC2P00	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3
ETSOC3P00	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3
ETSOU1P03	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3
ETSOV1F01	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3
ETSOV2F01	1,378	1,4988	1,6195	1,7402	1,861	1,9817	2,1024	2,2232	2,3439	2,4646	2,5854
ETSOV3F01	1,378	1,4988	1,6195	1,7402	1,861	1,9817	2,1024	2,2232	2,3439	2,4646	2,5854
ETSOV4F01	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3
ETSOV5F01	1,378	1,4988	1,6195	1,7402	1,861	1,9817	2,1024	2,2232	2,3439	2,4646	2,5854
ETSOV6F01	1,378	1,4988	1,6195	1,7402	1,861	1,9817	2,1024	2,2232	2,3439	2,4646	2,5854
ETWI25P00	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3
ETWI30P00	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	1,5	3

Table 51: Total Annual Min Capacity Investment, 2010-2020

Technology	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
ETGOCVP00	0	0	0	0	0	0,06	0,01	0,09	0	0	0
ETHYDMS01	0	0	0	0	0	0	0,0066	0	0	0	0
ETHYDMS02	0	0,097	0	0	0	0	0	0	0	0	0
ETHYDMS03	0,46	0	0	0	0	0,187	1,9371	0,7	4,05	0	0
ETSOU1P03	0,0001	0	0	0	0	0	0	0	0	0	0
ETWI30P00	0	0,03	0,051	0,0902	0	0,051	0	0	0	0	0

The above-mentioned technologies have a value of zero from 2020-2040. Remaining technologies are zero from 2010-2040.

Table 52: Total Technology Annual Activity Upper Limit, 2010-2040

Technology	
ETSOC1P00	82652,4
ETSOC2P00	82652,4
ETSOC3P00	82652,4
ETSOPV	97754,8814
ETSOV2F01	0
ETSOV3F01	0
ETSOV5F01	0
ETSOV6F01	0
ETWI25P00	53416,8
ETWI30P00	17939,2653

Table 53: Variable Cost, 2010-2040

Technology	
ETBACKSTOP	9999
ETBM00I00	2,25
ETBM00X00	1,5
ETCOSCP00	1,160944433
ETDSRCFI2	15,38888889
ETDSRCFR1	9,222222222
ETDSRCFU1	9,222222222
ETDSRCP03	0,555555556
ETEL00T00	0,00001
ETEL00TDI	0,00001
ETEL00TDR	0,00001
ETEL00TDU	0,00001
ETGOCVP00	0
ETHFGCP00	4,166666667
ETHYDMS01	2,03387915
ETHYDMS02	1,28647928
ETNGCCP00	0,659515521
ETNGGCP00	0,557627821
ETSOC1P00	0
ETSOC3P00	4,555555556
ETSOPV	0
ETSOU1P03	0
ETSOV1F01	0
ETSOV2F01	4,75
ETSOV3F01	5,277777778
ETSOV4F01	0

ETSOV5F01	4,75
ETSOV6F01	5,277777778
ETWI25P00	0
ETWI30P00	0

Table 54: Variable Cost, 2010-2019

Technology	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
ETBMCHP00	0,811	0,811	0,811	0,807	0,802	0,798	0,793	0,789	0,784	0,780
ETCO00I00	4,821	5,621	4,547	3,974	3,595	2,820	2,410	2,457	2,514	2,566
ETCO00X00	2,143	2,498	2,021	1,766	1,598	1,253	1,071	1,092	1,117	1,140
ETDS00I00	25,950	31,370	32,060	32,224	29,859	15,767	11,300	14,420	15,176	15,964
ETDS00X00	24,653	29,802	30,457	30,613	28,366	14,979	10,735	13,699	14,417	15,166
ETHF00I00	15,299	18,494	18,901	18,998	17,604	9,296	6,662	8,502	8,947	9,412
ETHF00X00	14,534	17,570	17,956	18,048	16,723	8,831	6,329	8,077	8,500	8,941
ETHYDMS03	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100	1,100
ETNG00I00	13,135	15,351	16,934	17,567	15,035	10,920	8,862	9,021	9,179	9,179
ETNG00X00	12,479	14,584	16,087	16,688	14,283	10,374	8,419	8,570	8,720	8,720
ETSOC2P00	0,683	0,683	0,683	0,655	0,628	0,601	0,573	0,546	0,519	0,492

Table 55: Variable Cost, 2020-2029

Technology	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
ETBMCHP00	0,775	0,774	0,772	0,771	0,769	0,768	0,766	0,765	0,763	0,762
ETCO00I00	2,623	2,680	2,737	2,794	2,851	2,908	2,936	2,965	2,993	3,021
ETCO00X00	1,166	1,191	1,216	1,242	1,267	1,293	1,305	1,318	1,330	1,343
ETDS00I00	16,818	17,705	18,658	19,643	20,662	21,779	22,301	22,824	23,347	23,869
ETDS00X00	15,978	16,820	17,725	18,661	19,629	20,690	21,186	21,683	22,179	22,676
ETHF00I00	9,915	10,438	11,000	11,581	12,181	12,840	13,148	13,456	13,764	14,072
ETHF00X00	9,420	9,916	10,450	11,002	11,572	12,198	12,490	12,783	13,076	13,369
ETHYDMS03	1,100	1,107	1,115	1,122	1,129	1,136	1,144	1,151	1,158	1,166
ETNG00I00	9,337	9,496	9,654	9,812	9,970	10,129	10,539	10,950	11,360	11,771
ETNG00X00	8,870	9,021	9,171	9,321	9,472	9,622	10,012	10,402	10,792	11,182
ETSOC2P00	0,464	0,464	0,464	0,464	0,464	0,464	0,464	0,464	0,464	0,464

Table 56: Variable Cost, 2030-2040

Technology	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040
ETBMCHP00	0,760	0,759	0,757	0,756	0,754	0,753	0,753	0,753	0,753	0,753	0,753
ETCO00I00	3,050	3,065	3,081	3,097	3,112	3,128	3,144	3,160	3,175	3,191	3,207
ETCO00X00	1,355	1,362	1,369	1,376	1,383	1,390	1,397	1,404	1,411	1,418	1,425
ETDS00I00	24,392	24,915	25,437	25,960	26,483	27,005	27,528	28,051	28,573	29,096	29,619
ETDS00X00	23,172	23,669	24,165	24,662	25,159	25,655	26,152	26,648	27,145	27,641	28,138
ETHF00I00	14,380	14,689	14,997	15,305	15,613	15,921	16,229	16,537	16,846	17,154	17,462
ETHF00X00	13,661	13,954	14,247	14,540	14,832	15,125	15,418	15,711	16,003	16,296	16,589
ETHYDMS03	1,173	1,180	1,187	1,195	1,202	1,209	1,209	1,209	1,209	1,209	1,209
ETNG00I00	12,182	12,524	12,866	13,208	13,550	13,893	14,235	14,577	14,919	15,261	15,603
ETNG00X00	11,573	11,898	12,223	12,548	12,873	13,198	13,523	13,848	14,173	14,498	14,823
ETSOC2P00	0,464	0,464	0,464	0,464	0,464	0,464	0,464	0,464	0,464	0,464	0,464

